

Facial Emotion Recognition under Occlusion: A Systematic Literature Review

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ABSTRACT

Facial Emotion Recognition (FER) faces significant challenges in real-world environments, particularly due to face occlusion and pose variations, which can obscure important features for expression recognition. This study offers a Systematic Literature Review (SLR) that includes 59 articles indexed in Scopus, Google Scholar, and Web of Science from Q1 to Q3, published between 2020 and 2025. The research utilises the Kitchenham methodology for the evaluation and examination of pertinent literature. The findings indicate that occlusion in the eye and mouth regions has a significant impact on the reduction of FER accuracy. Various approaches have been developed, such as the utilization of attention mechanisms in deep learning architectures and face reconstruction techniques, including frontalization, inpainting, and optical flow. While effective in specific contexts, numerous of these methodologies continue to face challenges in achieving optimal generalisation within the intricate conditions of the real world. These results underscore the necessity for further advancement and refinement of both methodologies and more representative and balanced datasets, in addition to the investigation of multimodal and real-time approaches to bolster the resilience of facial expression recognition systems against occlusion and fluctuating facial dynamics.

KEYWORDS

facial emotion, facial expression recognition, facial occlusion, occlusion, occluded face.

I. INTRODUCTION

FACIAL Expression Recognition (FER) is a rapidly advancing domain within image processing and computer vision. This technology has many important uses, such as making human-computer interaction more natural, creating security systems that use facial recognition, and analysing emotions in psychology and marketing [1]. Systems that can accurately read facial expressions can tell how someone is feeling in real time, which is very useful in many social and technological situations.

Facial expression recognition, however, encounters numerous intricate challenges in practice. Occlusion is a big problem for FER applications. It happens when other things, like glasses, masks, hair, hands, or other things that block the view of facial features, cover or block part of the face [2]. This occlusion is prevalent in real-world scenarios, especially in the post-pandemic context characterised by the extensive use of masks, rendering it a significant challenge that FER systems must tackle.

When the face is covered, important information is lost from the covered areas, which makes it harder for the system to get accurate expression features [3]. The recognition model must be able to work with incomplete data to still make accurate classifications when facial features that show certain emotions aren't there. This necessitates the creation of more resilient and flexible approaches for managing occlusion.

Different strategies have been devised to tackle the occlusion problem in FER. Image pre-processing techniques, such as detection and restoration of occluded areas [4], the use of 3D models to reconstruct the face [5], and machine learning algorithms that are

resistant to occluded facial data have been the main focus of research in recent years. These methods try to lessen the effects of occlusion and make it easier to recognise expressions in real life.

[6] recently came up with the Multi-Angle Feature Extraction (MAFE) method. This method improves the system's ability to deal with occlusion by combining global features at different scales, fine local features, and important features for specific regions. The Occlusion-RAFDB dataset had an accuracy of 89.42%, and the Occlusion-FERPlus dataset had an accuracy of 86.94%. This was better than what had been done before. This method also worked well on datasets that didn't have occlusion, showing that it is both flexible and reliable.

Moreover, a prevalent challenge associated with occlusion is considerable variation in facial pose. Prior research has introduced methodologies such as the Region Attention Network (RAN) [2] and the Patch Attention Convolutional Vision Transformer (PACVT) [7], which adeptly ascertain the significance of information from various facial regions. These methods effectively tackle both occlusion and pose variation, thereby improving expression recognition performance in intricate and dynamic environmental contexts.

The progress of deep learning and transformer technologies creates new possibilities for facial expression recognition that is resistant to occlusion. These models can learn more complex and contextual feature representations, which makes it easier to deal with missing data caused by occlusion [8]. Nevertheless, challenges such as the necessity for extensive datasets and computational resources persist as significant concerns.

Despite notable advancements in various methodologies, occlusion continues to be the principal challenge impacting FER system performance, especially in real-world scenarios characterised

by variability and disturbances [9]. Consequently, additional research and development are imperative to establish more resilient, adaptable, and precise systems capable of effectively managing occlusion and intricate environmental conditions.

This article aims to thoroughly examine the challenge of occlusion in facial expression recognition and to discuss various methods and recent advancements that seek to enhance the system's resilience to such conditions. This review is intended to be a significant reference for the advancement of more dependable and functional FER technology in the future. The research direction will be based on the following research questions (RQ):

- RQ1: What types of facial occlusion have a significant impact on the accuracy of FER systems?
- RQ2: What composition of data is needed to test the resilience of FER models against occluded face conditions?
- RQ3: What approaches have been implemented to enhance FER performance on occluded facial data?
- RQ4: How effective are existing approaches in improving FER accuracy on occluded facial data?

II. METHODOLOGY

This part gives an overview of the steps taken to do this systematic review, such as the sources of information, the search strategy, the criteria for article eligibility, the filtering methods, the coding, the data extraction, and the analysis procedures. We worked together throughout the process to make sure that each step was done correctly and without bias.

A. Search Strategy

We got the literature for this study by searching three big databases: Scopus, Google Scholar, and Web of Science (WoS). We used institutional accounts to get to these databases. To ensure the completeness of the search results, the article search was limited to publications from 2020 to 2025, with the aim of ensuring that the reviewed articles are recent and relevant research.

The search process utilized a search string designed with consideration of various synonymous terms and equivalents with similar meanings. This approach aimed to increase the coverage and relevance of the articles found. The search string used was as follows: ("occlusion" OR "occluded face" OR "face occlusion" OR "facial occlusion") AND ("facial expression recognition" OR "facial emotion recognition" OR "face emotion recognition"). A summary of the total number of papers retrieved can be seen in Table I.

TABLE I. NUMBER OF SEARCH RESULTS

Publisher	Result
Scopus	446
Google Scholar	199
Web of Science	64
Total	709

B. Inclusion and Exclusion Criteria

Inclusion and exclusion criteria are necessary to maintain the quality of the articles to be reviewed, prevent duplication of search results, and ensure the discussion remains focused on the predefined topic [10]. The criteria for article selection in this review are as follows:

- Articles must be from journals indexed in Q1, Q2, and Q3. Articles from journals indexed below Q3 will be excluded.

- Articles must meet two discussion requirements: first, they must include a discussion on face occlusion, and second, they must be applied to facial expression recognition or facial emotion recognition. Articles that meet only one of these two criteria will be excluded.

C. Study Selection

Table I shows the results of the search: Scopus found 446 articles, Google Scholar Direct found 199, and WoS found 64. This is a total of 709 articles from all three sources. The article selection process is illustrated using a PRISMA diagram in Figure 1. In a systematic review, PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is a diagram that visually represents the study selection process, which includes identification, screening, eligibility, and inclusion/exclusion of studies [11].

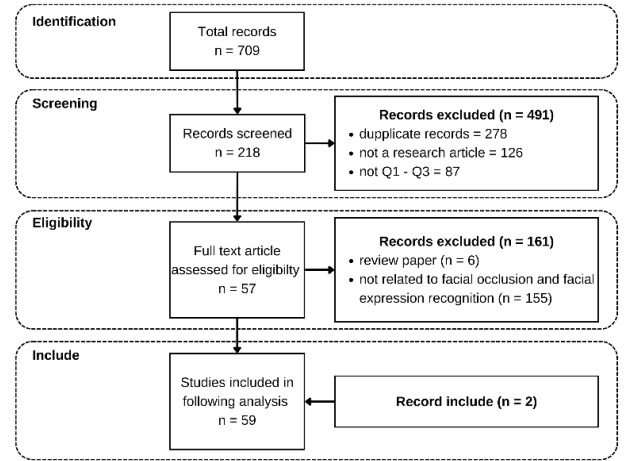


Fig. 1. Review Protocol.

D. Selection Result

A total of 59 articles were included in this review after undergoing a rigorous selection process. All selected articles are publications indexed in the databases listed in Table I, with quartile categories ranging from Q1 to Q3, ensuring the quality and relevance of the analyzed research. A complete list of these articles, including their publishers and quartile rankings, is presented in Table II.

TABLE II. FINAL ARTICLE RESULT

Quartile	Papers	Sum
Q1	[5], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [7], [40], [41], [42], [43], [2], [4], [44], [45], [46], [47], [48], [49], [50]	43
Q2	[51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65]	15
Q3	[66]	1

III. RESULT

A. RQ1: What types of facial occlusion have a significant impact on the accuracy of FER systems?

To address Research Question 1 (RQ1) concerning the types of facial occlusion that substantially affect the accuracy of FER systems, a review was performed on the various types of occlusions examined in the literature. However, based on the analysis of the articles that

met the selection criteria, most studies did not explicitly mention the specific types of occlusion addressed.

Generally, occlusion is mentioned broadly as a disturbance to facial expression detection, without classifying its form or location. Nevertheless, some studies focus on specific types of occlusion, such as those induced by variations in head pose, glasses, medical masks, or hair, and strive to create methods suited to these circumstances. Table III shows the different types of occlusion that this study looked at. Based on Table III, the types of occlusion used by researchers to test their proposed methods can be categorized into two types: natural occlusion and synthetic occlusion.

TABLE III. TYPE OF OCCLUSIONS DATA

Occlusion	Type of Occlusion Data	Paper
Head Pose Variation	Natural	[5], [18], [22], [54], [57], [61], [62]
Illumination Variation	Natural	[18]
Occlusion in NFPA Context	Natural	[42]
Medical Mask	Synthetic	[14], [17], [66]
Area Around The Eyes and Hair	Synthetic	[16], [21]
Block Masking	Synthetic	[4], [43]
Hair and Glasses	Natural	[18], [55], [59]

Natural Occlusion: Occlusion that occurs naturally in real-world environments and is generally uncontrollable by the system, such as head pose variation, illumination variation, and other attributes like hair and glasses. Studies [5], [18], [22], [54], [57], [61], and [62] show that changes in head pose can significantly obscure important areas on the face, thereby reducing the accuracy of FER systems. A similar finding was observed in illumination variation [18], which can lead to inconsistencies in visual features, as well as personal attributes like hair and glasses [55], [18], which cover key facial areas such as the eyes and eyebrows.

Synthetic Occlusion: Occlusion that is artificially simulated in experiments to assess the resilience of FER systems under extreme conditions. Common examples include medical masks [14], [17], [66], which cover the mouth and chin area, block masking [43], [4], which involves creating black block objects to randomly cover facial areas, and occlusion of specific areas like the eyes and hair [21], [16]. This type of occlusion is designed to isolate the impact of certain areas on FER accuracy, helping to analyze the system's sensitivity to the loss of visual information.

From this synthesis, it can be concluded that occlusions covering key expression areas such as the eyes, mouth, and eyebrows have a significant impact on the accuracy of FER systems. Both natural and synthetic occlusions have been shown to reduce performance, depending on which part of the face is covered and the overall visual context. Therefore, mapping the types of occlusion is a crucial first step in developing FER systems that are robust to real-world conditions.

B. RQ2: What composition of data is needed to test the resilience of FER models against occluded face conditions?

To answer RQ2, this study finds datasets that have facial conditions that are blocked, both naturally and artificially. These datasets are essential for assessing the resilience of Facial Expression Recognition (FER) models to the degradation of visual information in particular

facial regions. Several datasets have been explicitly designed or modified to reflect various occlusion scenarios, including the use of medical masks, partial face covering, and visual disturbances caused by viewpoint angles or facial attributes such as hair and glasses. This section will talk about different datasets that are often used in research to see how well FER models do when they have to deal with occlusion problems. It will also talk about how each dataset is different in terms of how well it represents different types of occlusion.

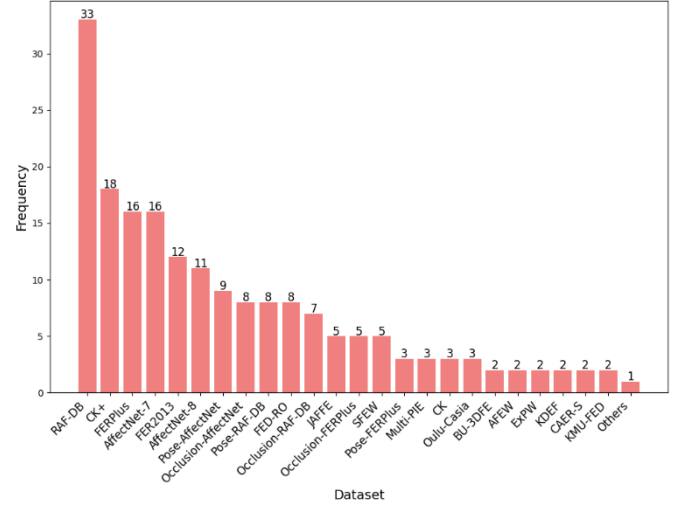


Fig. 2. Dataset freq used.

The graph shows the frequency distribution of dataset usage, which shows that many popular and specialised datasets are often used for this purpose in research. Datasets like RAF-DB [67], CK+ [68], AffectNet-7 [69], FERPlus [70], and FER2013 [71] are the most used.

- CK+ (Extended Cohn-Kanade) is a dataset that has 593 sequences of facial expressions from 123 people. Each sequence has an emotion label for the last frame. This dataset is widely used for dynamic expression analysis because it focusses on controlled expressions that change from neutral to peak.
- AffectNet-7 is a part of AffectNet and has more than 450,000 facial images with labels for 7 basic emotions. The images were collected through internet searches utilising emotion-related keywords and were manually annotated by human annotators, resulting in one of the largest and most intricate datasets in emotion recognition.
- FER2013 is a dataset introduced in the Kaggle competition, which is a dataset of 35,887 greyscale facial images that are 48x48 pixels in size and have labels for seven basic emotions. The pictures were automatically taken from the web and became an early standard for recognising facial expressions in a simple visual domain.
- FERPlus is an improved version of FER2013. This dataset also contains 35,887 facial images of size 48x48 pixels, but the emotion labels were updated through re-annotation by eight raters per image, covering 8 emotions: happy, sad, surprise, fear, disgust, anger, neutral, and contempt.

Additionally, there are datasets specifically designed to test models under occlusion conditions, such as Occlusion-AffectNet, Occlusion-RAF-DB, Occlusion-FERPlus, Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus. All these datasets were created by Wang, K. et al. (2020) [2], with the goal of testing the resilience of models against facial data variations in pose and various forms of occlusion.

- Occlusion-AffectNet is a version of AffectNet that has been modified by adding artificial occlusion (such as hands, masks, or other objects covering part of the face) to evaluate the resilience of models against partially occluded facial expressions.
- Occlusion-RAF-DB is a variant of RAF-DB that introduces occlusion to facial images, either synthetically or from original images with parts of the face covered (e.g., by hands or glasses). This version is used to test the robustness of models under real-world, non-ideal conditions.
- Occlusion-FERPlus is a modification of FERPlus where facial images are occluded in areas such as the eyes, nose, or mouth, aiming to assess the model's ability to recognize emotions when important facial features are obstructed.
- Pose-AffectNet is a subset of AffectNet focused on facial pose variations, including images with sideways, downward, upward, or tilted poses. This dataset is designed to evaluate the performance of models in recognizing expressions from non-frontal face viewpoints.
- Pose-RAF-DB is a version of RAF-DB that focuses on facial pose variations, including both frontal and non-frontal poses. Pose labels are typically obtained from landmarks or face pose estimation to analyze how facial angle influences emotion classification accuracy.
- Pose-FERPlus is a dataset resulting from the augmentation or subset of FERPlus that retains facial pose variations, used to test model reliability in recognizing expressions when the face is not directly facing the camera.

Information about these datasets is crucial for answering RQ2, as this study provides facial data with real occlusion disturbances,

allowing the evaluation of FER model performance in more challenging and realistic situations.

Meanwhile, datasets with lower frequencies such as JAFFE [72], SFEW [73], and Oulu-Casia [74] still provide valuable contributions to model testing, especially for specific conditions or expressions. In Figure 2, there is a category labeled "Others," which represents datasets in this study with a frequency of one, and these will still be presented in Table IV as alternative datasets that can be used to test FER models under occlusion conditions.

TABLE IV. THE "OTHERS" CATEGORY DATASET AS SHOWN IN FIGURE 2

Dataset	
SAMM [75]	LFW [76]
RafD [77]	GENKI [78]
RM-FRD [79]	DFEW [80]
OLFER [36]	CeleBA [81]
OLFED-HO [44]	Casia-WebFace [82]
NFPA [42]	CFEE [83]
MYRAF-3_KN95 [66]	CEFE [30]
MYLFW_KN95 [66]	CASME II [84]
MMI [85]	Aff-Wild2 [86]

The frequency data of dataset usage not only mirrors the inclinations of the research community but also offers a definitive summary of suitable and illustrative dataset selections for evaluating the robustness of FER models in the presence of occluded data. Researchers can utilise this information as a reference for selecting datasets that correspond with their testing requirements, whether they are general or specifically focused on occlusion, thus improving the validity and robustness of the models under development.

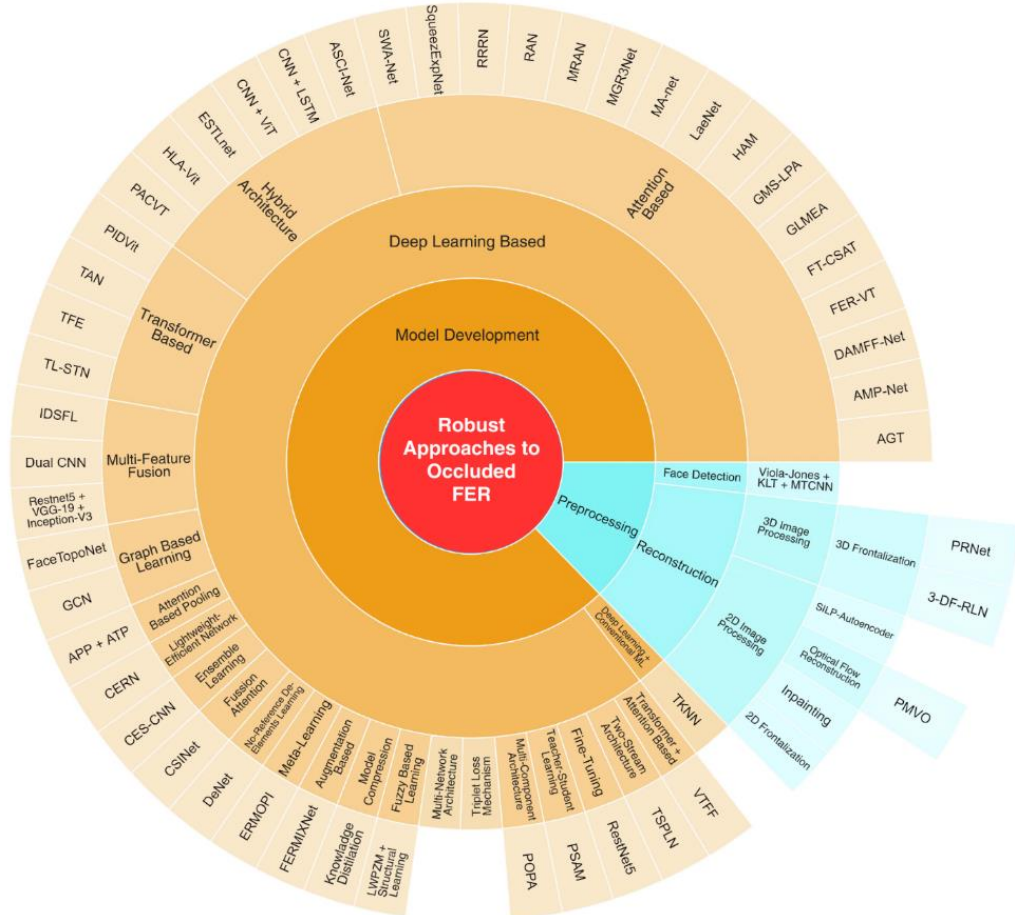


Fig. 3. Categories of Approaches to Address Occlusion Issues in FER.

C. RQ3: What approaches have been implemented to enhance FER performance on occluded facial data?

All of the chosen papers that talk about occlusion in FER suggest different ways to solve this problem. These differences show that researchers use different strategies to make FER work better on occluded facial data. In this section, we try to group these methods into groups based on their main features, like how they preprocess data, how they build models, or how they extract features. To provide a clearer picture, we present a visualization (Figure 3) that shows the distribution of the approach categories used in these studies.

The research articles from the period 2020-2025 presented in Figure 3 indicate that model development approaches based on deep learning dominate the methods used to tackle occlusion in FER. A more detailed look at the different ways to develop deep learning-based models shows that the attention mechanism is the most popular one. The attention mechanism is a technique that lets the model dynamically focus on the most informative or nonoccluded parts of the face when it is trying to recognise facial expressions. In the realm of FER, attention is employed to ignore occluded or extraneous facial components (e.g., obscured by hands, masks, or other objects) [34], or to highlight essential expressive characteristics such as the eyes, eyebrows, or mouth [32].

In addition to the model development-focused approach, there are also results indicating other approaches such as preprocessing techniques or feature extraction processes. In the preprocessing stage, techniques such as 3D [5] and 2D [43] face reconstruction dominate. Face reconstruction with frontalization is also used to manipulate head poses that are not facing forward to make them face the front, either by using half-face data [62] or by combining two rotated images [61]. This study also highlights a feature extraction approach, specifically through multi-channel feature fusion [34].

D. RQ4: How effective are existing approaches in improving FER accuracy on occluded facial data?

To answer RQ4 regarding the improvement in accuracy of existing methods in occluded FER, this study applied a selection rule where the included papers are those that perform tests comparing the baseline condition and the results of the proposed method to assess its impact on FER system accuracy. Unfortunately, not all papers included in the study provided comparisons between the baseline condition and the results of their proposed method. The papers that meet the inclusion criteria are presented in Table V.

A total of 30 studies specifically present the results of ablation experiments, comparing the performance evaluation of models under baseline conditions (without additional components or modifications) and the proposed method (with the suggested approach or architecture). Each study reports evaluation metrics such as accuracy (and, in some cases, weighted average recall) to demonstrate the performance improvement of the model across various benchmark datasets such as RAF-DB, AffectNet, FER2013, CK+, FED-RO, and others.

In the papers presented in Table V, the experiments are not always conducted on just one dataset but may involve two or more datasets. However, in this study, we focus on ablation experiment data from datasets that are representative of occluded facial data and are commonly used to validate the proposed methods, such as RAF-DB, AffectNet, and FED-RO, which contain extensive facial occlusion data

The results of ablation experiments are crucial in measuring the

effectiveness of the contributions from the approaches taken to address the occlusion problem in FER. These results also serve as important insights to answer RQ4 regarding the effectiveness of the approaches implemented.

TABLE V. COMPARISON OF PERFORMANCE BETWEEN BASELINE CONDITION (A) AND PROPOSED METHOD RESULT (B)

Method	Evaluation Metrics	Performance %		Dataset
		A	B	
Resnet50 [14]	Accuracy	72.00	78.00	M-LFW-FER
CERN [15]	Accuracy	63.25	65.25	FED-RO
CSINet [53]	Accuracy	76.43	89.67	RAF-DB
GMS-LPA [54]	Accuracy	84.14	87.02	Multi-PIE
Improved Efficient MAS [19]	Accuracy	84.63	86.51	RAF-DB
ESTLnet [20]	Accuracy	78.54	89.38	Oulu-Casia
FT-CSAT [22]	Accuracy	87.29	88.61	RAF-DB
HLA-Vit [57]	Accuracy	87.22	90.45	RAF-DB
TL-STN [24]	Accuracy	96.87	99.41	CK+
SWA-Net [26]	Accuracy	87.13	90.03	RAF-DB
FER-VT [27]	Accuracy	76.83	84.31	RAF-DB
VTFF [28]	Accuracy	86.37	88.19	RAF-DB
ASCI-Net [60]	Accuracy	86.29	86.86	RAF-DB
FERMIXNet [29]	Accuracy	82.99	86.67	Occlusion-RAF-DB
Face Frontalization [61]	Accuracy	70.95	73.17	FER2013
GLMEA [30]	Accuracy	88.04	90.09	RAF-DB
HAM [31]	Accuracy	57.80	61.04	Pose-AffectNet
MA-net [33]	Accuracy	53.55	55.30	Pose-AffectNet
IDFSL [34]	Accuracy	52.69	64.02	Occlusion-AffectNet
Knowledge Distillation [36]	Accuracy	77.50	79.50	RAF-DB
MGR ³ Net [37]	Accuracy	85.23	88.82	RAF-DB
MRAN [38]	Accuracy	87.84	90.03	RAF-DB
DAMFF-Net [64]	Accuracy	80.15	85.94	RAF-DB
PACVT [7]	Accuracy	84.96	88.21	RAF-DB
RAN [2]	Accuracy	73.33	83.63	Occlusion-FERPlus & Pose-FERPlus
RAN [44]	Accuracy	84.80	89.00	AffectNet
TAN [46]	Accuracy	86.62	89.12	RAF-DB
DeNet [49]	Accuracy	69.75	71.50	FED-RO
APP + ATP [50]	Accuracy	85.02	86.03	Aff-Wild2
RRRN [39]	Weight Avarage Recall	0.487	0.557	SAMM + CASME II

IV. GENERAL DISCUSSION

A. Significant Impact of Occlusion on FER

Based on the findings in this study, the types of occlusion identified in Table III show a significant contribution to the decline in the accuracy of FER systems. These types of occlusions can be categorized into three main groups based on the areas of the face that are covered: the eye area, the nose area, and the mouth area. Occlusion in the eye area tends to disrupt the identification of expressions that rely on the dynamics of eyebrow and eyelid movement, while occlusion around the nose, although having a relatively smaller impact, still affects the overall facial feature structure. Meanwhile,

occlusion in the mouth area significantly affects emotion recognition, particularly expressions associated with mouth shape changes, such as happiness and sadness. This classification provides an important foundation for developing more robust FER methods that are resilient to various types of facial occlusions.

Occlusion in the eye area in real-world environments is commonly caused by the use of glasses, both for medical or protective purposes, as well as by hair covering parts of the face [59], such as bangs or naturally falling strands. This condition is frequently encountered in various surveillance system applications, especially in the context of monitoring vehicle drivers or Advanced Driver Assistance Systems (ADAS) [63], where cameras are directed at the driver's face to detect alertness or emotions. Such occlusion can hinder the accuracy of systems in recognizing facial expressions, as the eye area plays a crucial role in expressing emotions like anger, fear, and surprise. Therefore, the presence of occlusion in the eye area presents a unique challenge in developing vision-based FER systems that are implemented in real-world scenarios.

Occlusion in the mouth area also significantly impacts the accuracy of FER systems. Usually, occlusion in the mouth area can take the form of facial accessories such as masks, which are widely used by people during illness or as part of regular habits. During the COVID-19 pandemic, people were required to wear masks, causing FER systems to become less accurate [14]. Another form of occlusion is when the hand covers the mouth or face, which often occurs during online learning sessions [44]. This situation arises when students feel bored or due to health factors such as a stiff neck or other medical issues [87].

Occlusion in the nose area also influences the recognition of several basic emotions, although its contribution is not as significant as occlusion in the eye or mouth areas. Previous studies have shown that approaches based on modifying muscle movements around the nose area can improve confidence levels in detecting basic emotions like happiness, sadness, and anger [62]. This suggests that, although the nose area is not the primary center of facial expression, its role is still significant in supporting the overall representation of emotional expressions. Therefore, considering the muscle characteristics in this area could be an important aspect of developing more accurate and robust FER systems against partial occlusion.

In addition to occlusion, another factor contributing to the decrease in FER system accuracy is head pose variation [5], [18], [22], [54], [57], [61], [62], which occurs when the orientation of the face in the image is not directly facing the camera due to changes in head direction or tilt. This variation can cause spatial distortion in facial features that are key to expression recognition, making it difficult for the model to consistently extract emotional information. Pose changes such as yaw, pitch, or roll rotations can obscure or even hide essential areas of the face, such as the eyes, eyebrows, and mouth, which significantly impact expression classification. Therefore, handling head pose variation becomes an important challenge in developing robust and reliable FER systems in real-world environments.

B. Datasets For Testing The Resilience Of FER Models Against Occluded Facial Data

In efforts to build robust models against occluded facial data, a collection of facial datasets representing real-world conditions, both with and without scenario modifications, is essential. Ideally, these datasets should include variations of occlusion caused by objects such as hands, hair, masks, or other environmental elements, as well as other realistic conditions such as poor lighting, extreme poses, and complex facial expressions. The availability of data with these

characteristics is crucial for both training the model to improve its generalization ability and for validating the model's performance in real-world situations more accurately.

Facial data can be gathered from several public datasets such as FED-RO, RAF-DB, AffectNet, FER2013, FERPlus, and others that include a variety of occlusion types. Occlusion types can also be synthetically generated, such as by positioning random black blocks that cover portions of the face [4], [43]. Datasets specifically collected with certain types of occlusion include Occlusion-AffectNet, Occlusion-RAF-DB, and Occlusion-FERPlus for datasets with diverse occlusion types, and Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus for facial data with varying head poses and angles, which are subsets of AffectNet, RAF-DB, and FERPlus.

The class distribution balance in a dataset is a critical factor in developing reliable and accurate machine learning models. Data imbalance, such as the dominance of certain classes, can lead to models being biased towards the majority class while neglecting the representation of minority classes. Therefore, datasets with balanced label distributions are essential to ensure that the built model can recognize all categories proportionally and maintain generalization performance on unseen data.

- FERPlus is an extension of the FER2013 dataset that includes 28,709 training images and 3,589 test images. This dataset was re-annotated by ten annotators per image, resulting in higher-quality and more reliable labels compared to the original FER2013 dataset [2].
- CK+ (Extended Cohn-Kanade) is a commonly used dataset for FER tasks in controlled laboratory conditions. Although it was not explicitly designed with class balance in mind, its smaller size and controlled data collection environment make it highly consistent and clean [38].
- RAF-DB is an in-the-wild dataset containing both basic and compound expressions. The basic expression subset consists of 15,339 images split into training and testing data. Since the data was collected from real-world environments, RAF-DB offers high variability in terms of pose, lighting, and occlusion [38].
- FED-RO is a facial expression dataset containing real occlusion data collected through internet searches. This dataset is focused on testing and designed to avoid overlap with other datasets like RAF-DB or AffectNet, making it useful for evaluating model robustness against occlusion in real-world environments [38].
- AffectNet-7 is a subset of the large AffectNet dataset, with over 1 million facial images collected from the internet. Approximately 400,000 images were manually annotated into seven basic emotion categories. Its large size and diverse image collection conditions make AffectNet-7 one of the most representative datasets for real-world FER applications [38].
- FER2013 was developed for the ICML 2013 competition and consists of 48×48 pixel grayscale images. Compared to AffectNet and RAF-DB, FER2013 has a smaller scope and more limited expression variety, but it is still widely used as a benchmark for FER model testing [2].
- Occlusion-FERPlus, Occlusion-AffectNet, and Occlusion-RAF-DB are modified subsets of the main datasets that explicitly add occlusion elements (both natural and artificial) to the facial images. These datasets are intended to provide focused evaluation on model robustness in recognizing emotions from occluded faces [37] [28].
- Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus are subsets designed to evaluate model performance under facial pose variations. Additional annotations are made to identify images

with rotations greater than 30° or 45°, allowing for in-depth study of model robustness against changes in face orientation [46] [28].

In terms of data balance, AffectNet-7 is considered one of the most comprehensive and diverse datasets, both in terms of data quantity and annotation variations. This makes it highly suitable for developing FER models that can generalize effectively in real-world environments. For validation of models on occluded facial data with varying head poses, datasets like Occlusion-FERPlus, Occlusion-AffectNet, Occlusion-RAF-DB, or Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus can be used.

C. Technical Approaches to Address Occlusion Issues In FER

In this study, the approach taken by researchers is predominantly focused on model development aimed at building robust models against occluded data based on deep learning (Fig. 3). When broken down in more detail, the dominant approach is based on attention mechanisms, which assign different weights to occluded parts of the face and give more importance to non-occluded areas.

Attention mechanisms have played a significant role in facial expression recognition, both in static and dynamic contexts. At the spatial level, both global and local attention are used to identify key areas of the face that are closely related to expressions by designing spatial attention modules that highlight relevant information from the entire face as well as specific local regions [38]. In dynamic expression recognition, attention mechanisms are used to enhance the discriminative power of spatial-temporal features by utilizing transformer architectures, enabling the model to understand expression sequences more effectively [20].

Additionally, attention mechanisms can also be applied within Convolutional Neural Network (CNN) architectures to develop FER systems that are robust against occlusion, with gating units directing the focus from occluded areas to exposed sections of the face [18]. One innovative approach is the Patch-Gated Convolutional Neural Network (PG-CNN), which is designed to maintain focus on non-occluded, discriminative parts of the face while processing information from occluded areas, thus improving feature extraction effectiveness in complex visual conditions [17].

In addition to the model development-focused approach, another focus area in the machine learning/deep learning pipeline is preprocessing. The preprocessing techniques in this study focus on face reconstruction, both using optical [16], [63], or techniques for frontalizing the) [61], [62].

Frontalization techniques can be performed in both 2D and 3D image processing. In 2D image processing, frontalization is carried out by augmenting a sideways-facing image of the face into a frame facing the opposite direction, then blending the original and augmented images together using blending techniques to create a frame that appears to face directly forward [61]. In 3D image processing, frontalization involves converting the 2D image into 3D, adjusting the face's position to face forward, and then converting it back into 2D. This approach can be implemented using PRNet [62].

Another form of face reconstruction can be achieved using Generative Adversarial Networks (GANs). GANs can restore occluded parts of the face by detecting and removing the occluding object and filling in the missing section with a representative portion from the available database, commonly referred to as inpainting [43].

The approach to addressing occlusion in FER with a focus on preprocessing is considered crucial in achieving significant results, especially with frontalization techniques. This is based on the idea that frontal faces have better visibility of key emotional determinants, particularly the eyes, facial muscles, and mouth [88]. This ensures

that the model is not biased towards the data being tested, especially when working with models trained on datasets that lack sufficient data on occluded faces or head pose variations. This approach helps overcome model bias caused by the limited variation of occluded faces in the training dataset.

V. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

Although various technical approaches have been discussed in this study to address the challenges of occlusion in facial expression recognition, there are several limitations that need to be considered and serve as avenues for future research development, including:

A. Challenges in Dataset Representation

There are datasets that include occluded faces and head pose variations, but most of the data still comes from real-world situations, which don't always show systematic occlusion. For instance, hands can block parts of the face in certain situations, like when someone is learning online. Also, the sample sizes are not evenly spread out among expression classes and types of occlusion, which can have a big effect on how well the model works. We need projects to make new datasets that clearly show different types of occlusion, with more detailed notes about things like the type of object that is blocking the view, the facial areas that are blocked, and the situation (like what activities are going on or how someone is feeling).

B. Reliance Only on Attention Mechanism and CNN

Most current methods rely heavily on attention mechanisms in CNN or transformer architectures. These methods work well, but they can have problems when there is a lot of occlusion or visual noise (like bad lighting or low resolution). Using a combination of audio, text (like captions or transcripts), and physiological signals (like EEG or EDA) can help people understand their emotions better when they can't see things clearly because of occlusion.

C. Model Generalization to Real-World Data

A lot of models do well on test data from the same dataset (intra-dataset testing), but do poorly on test data from different datasets (cross-dataset testing). This shows that models have a problem with generalisation. We need to develop domain adaptation or domain generalisation methods to make sure that models can handle differences in data distribution between domains, such as occlusion, pose, and expression conditions.

D. Challenges in Preprocessing Techniques

Frontalization and inpainting are examples of reconstruction techniques that are often used in preprocessing. These techniques often make fake results that aren't perfect. Optical flow and other techniques need sequential data (like videos), which isn't always easy to find. We need to come up with more realistic and situation-based reconstruction methods, like hybrid generative models that use both GANs and diffusion models, and methods that use more accurate 3D face priors.

E. Limited Evaluation in Real-World Contextual Conditions

Most evaluation studies remain confined to controlled or semi-controlled environments, inadequately reflecting the intricacies of real-world applications such as e-learning, emotional monitoring in smart classrooms, or Advanced Driver Assistance Systems (ADAS). Future research must assess FER systems in authentic deployment contexts, including educational, occupational, or digital social interactions, while taking into account environmental variables and user behavior.

VI. CONCLUSION

This study presents a systematic literature review (SLR) on Facial Emotion Recognition (FER) under occluded facial conditions, covering literature from 2020 to 2025. The review includes 59 articles from reputable journals (Q1–Q3), examining technical strategies, dataset challenges, and the effectiveness of proposed approaches.

The study found several key insights regarding occlusion in FER. Occlusion in the eye and mouth areas has the most significant impact on the accuracy of FER systems. Both natural occlusions, such as glasses, hair, and head pose, and synthetic occlusions, such as random masking, significantly affect FER performance, especially when they cover key facial expression areas. The review also highlighted popular datasets used to test model robustness against occlusion, such as Occlusion-AffectNet, Occlusion-RAF-DB, and Pose-FERPlus. However, many of these datasets do not fully represent real-world conditions or explicitly include occlusion-type annotations, and class imbalance and lack of contextual representation remain critical challenges.

The review also examined the technical approaches used to tackle occlusion in FER, with a dominant focus on deep learning models integrated with attention mechanisms, both spatial and temporal. Additionally, preprocessing techniques, such as face reconstruction, frontalization, optical flow, and inpainting with GANs, were emphasized to improve the model's ability to focus on non-occluded areas of the face and improve accuracy despite occlusion. The majority of the 30 studies that compared baseline and proposed methods demonstrated a significant improvement in accuracy, with gains of 5–15%, depending on the dataset used, which demonstrates that approaches specifically designed to address occlusion have a positive effect on system performance.

In conclusion, the study emphasizes that, although many technical approaches have been developed to overcome the challenges of occlusion in FER, there are limitations in model generalization, dataset representation, and real-world application scenarios. Future research should focus on the development and optimization of both technical strategies and datasets with detailed occlusion annotations, integrating multimodal inputs such as visual, audio, and physiological data to support FER in extreme occlusion conditions, and creating end-to-end systems capable of handling occlusion across different domains. This SLR is expected to be an important reference for researchers and practitioners aiming to develop FER systems that are not only technically accurate but also robust against visual disturbances and applicable in real-world scenarios like e-learning, smart vehicles, and human-computer interaction systems.

DECLARATION OF CONFLICTS OF INTEREST

No conflict of interest exists, as the authors affirm that there are no circumstances that could have influenced the content, interpretation, or conclusions of this article, and that all aspects of the research were conducted with full academic integrity and transparency.

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REFERENCES

- [1] O. S. Ekundayo and S. Viriri, "Facial Expression Recognition: A Review of Trends and Techniques," *IEEE Access*, vol. 9, pp. 136944–136973, 2021, doi: 10.1109/ACCESS.2021.3113464.
- [2] K. Wang, X. Peng, J. Yang, D. Meng, and Y. Qiao, "Region Attention Networks for Pose and Occlusion Robust Facial Expression Recognition," *IEEE Trans. Image Process.*, vol. 29, pp. 4057–4069, 2020, doi: 10.1109/TIP.2019.2956143.
- [3] A. F. Abate, L. Cimmino, B. C. Mocanu, F. Narducci, and F. Pop, "The limitations for expression recognition in computer vision introduced by facial masks," *Multimed. Tools Appl.*, vol. 82, no. 8, pp. 11305–11319, Mar. 2023, doi: 10.1007/s11042-022-13559-8.
- [4] D. Sun, W. Xie, Z. Ding, and J. Tang, "SILP-autoencoder for face de-occlusion," *Neurocomputing*, vol. 485, pp. 47–56, 2022, doi: 10.1016/j.neucom.2022.02.035.
- [5] N. Sun, J. Tao, J. Liu, H. Sun, and G. Han, "3-D Facial Feature Reconstruction and Learning Network for Facial Expression Recognition in the Wild," *IEEE Trans. Cogn. Dev. Syst.*, vol. 15, no. 1, pp. 298–309, 2023, doi: 10.1109/TCDS.2022.3157772.
- [6] Y. Li, H. Liu, J. Liang, and D. Jiang, "Occlusion-Robust Facial Expression Recognition Based on Multi-Angle Feature Extraction," *Appl. Sci.*, vol. 15, no. 9, p. 5139, May 2025, doi: 10.3390/app15095139.
- [7] C. Liu, K. Hirota, and Y. Dai, "Patch attention convolutional vision transformer for facial expression recognition with occlusion," *Inf. Sci. (Nij.)*, vol. 619, no. 5, pp. 781–794, 2023, doi: 10.1016/j.ins.2022.11.068.
- [8] T. Kopalidis, V. Solachidis, N. Vretos, and P. Daras, "Advances in Facial Expression Recognition: A Survey of Methods, Benchmarks, Models, and Datasets," *Inf.*, vol. 15, no. 3, 2024, doi: 10.3390/info15030135.
- [9] D. Zeng, R. Veldhuis, and L. Spreeuwers, "A survey of face recognition techniques under occlusion," *IET Biometrics*, vol. 10, no. 6, pp. 581–606, 2021, doi: 10.1049/bme2.12029.
- [10] A. Falavigna and M. Blauth, "Critical review of a scientific manuscript: a practical guide for reviewers," *J. Neurosurg.*, vol. 128, no. January, pp. 312–321, 2018, doi: 10.3171/2017.5.JNS17809.312.
- [11] M. J. Page *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, 2021, doi: 10.1136/bmj.n71.
- [12] H. Liu, H. Cai, Q. Lin, X. Li, and H. Xiao, "Adaptive Multilayer Perceptual Attention Network for Facial Expression Recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 9, pp. 6253–6266, 2022, doi: 10.1109/TCSVT.2022.3165321.
- [13] N. Sun, Y. Song, J. Liu, L. Chai, and H. Sun, "Appearance and geometry transformer for facial expression recognition in the wild," *Comput. Electr. Eng.*, vol. 107, 2023, doi: 10.1016/j.compeleceng.2023.108583.
- [14] G. Castellano, B. De Carolis, and N. Macchiarulo, "Automatic facial emotion recognition at the COVID-19 pandemic time," *Multimed. Tools Appl.*, vol. 82, no. 9, pp. 12751–12769, 2023, doi: 10.1007/s11042-022-14050-0.
- [15] D. Gera, S. Balasubramanian, and A. Jami, "CERN: Compact facial expression recognition net," *Pattern Recognit. Lett.*, vol. 155, pp. 9–18, 2022, doi: 10.1016/j.patrec.2022.01.013.
- [16] D. Poux, B. Allaert, N. Ihaddadene, I. M. Bilasco, C. Djeraba, and M. Bennamoun, "Dynamic Facial Expression Recognition under Partial Occlusion with Optical Flow Reconstruction," *IEEE Trans. Image Process.*, vol. 31, pp. 446–457, 2022, doi: 10.1109/TIP.2021.3129120.
- [17] R. Khoeun, P. Chopuk, and K. Chinnasarn, "Emotion Recognition for Partial Faces Using a Feature Vector Technique," *Sensors*, vol. 22, no. 12, 2022, doi: 10.3390/s22124633.
- [18] S. Kuruvayil and S. Palaniswamy, "Emotion recognition from facial images with simultaneous occlusion, pose and illumination variations using meta-learning," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 9, pp. 7271–7282, 2022, doi: 10.1016/j.jksuci.2021.06.012.
- [19] N. Li, Y. Huang, Z. Wang, Z. Fan, X. Li, and Z. Xiao, "Enhanced Hybrid Vision Transformer with Multi-Scale Feature Integration and Patch Dropping for Facial Expression Recognition," *Sensors*, vol. 24, no. 13, p. 4153, Jun. 2024, doi: 10.3390/s24134153.
- [20] W. Gong, Y. Qian, W. Zhou, and H. Leng, "Enhanced spatial-temporal learning network for dynamic facial expression recognition," *Biomed. Signal Process. Control*, vol. 88, no. PC, p. 105316, 2024, doi: 10.1016/j.bspc.2023.105316.

- [21] M. Kolahdouzi, A. Sepas-Moghaddam, and A. Etemad, "FaceTopoNet: Facial Expression Recognition Using Face Topology Learning," *IEEE Trans. Artif. Intell.*, vol. 4, no. 6, pp. 1526–1539, Dec. 2023, doi: 10.1109/TAI.2022.3207450.
- [22] H. Yao, X. Yang, D. Chen, Z. Wang, and Y. Tian, "Facial Expression Recognition Based on Fine-Tuned Channel-Spatial Attention Transformer," *Sensors*, vol. 23, no. 15, p. 6799, Jul. 2023, doi: 10.3390/s23156799.
- [23] H. Kim, J. H. Lee, and B. C. Ko, "Facial Expression Recognition in the Wild Using Face Graph and Attention," *IEEE Access*, vol. 11, no. June, pp. 59774–59787, 2023, doi: 10.1109/ACCESS.2023.3286547.
- [24] J. Kim and D. Lee, "Facial Expression Recognition Robust to Occlusion and to Intra-Similarity Problem Using Relevant Subsampling," *Sensors*, vol. 23, no. 5, 2023, doi: 10.3390/s23052619.
- [25] M. Ahmady, S. S. Mirkamali, B. Pahlevanzadeh, E. Pashaei, A. A. R. Hosseiniabadi, and A. Slowik, "Facial expression recognition using fuzzified Pseudo Zernike Moments and structural features," *Fuzzy Sets Syst.*, vol. 443, pp. 155–172, 2022, doi: 10.1016/j.fss.2022.03.013.
- [26] S. Qiu, G. Zhao, X. Li, and X. Wang, "Facial Expression Recognition Using Local Sliding Window Attention," *Sensors*, vol. 23, no. 7, 2023, doi: 10.3390/s23073424.
- [27] Q. Huang, C. Huang, X. Wang, and F. Jiang, "Facial expression recognition with grid-wise attention and visual transformer," *Inf. Sci. (Nyl.)*, vol. 580, pp. 35–54, 2021, doi: 10.1016/j.ins.2021.08.043.
- [28] F. Ma, B. Sun, and S. Li, "Facial Expression Recognition With Visual Transformers and Attentional Selective Fusion," *IEEE Trans. Affect. Comput.*, vol. 14, no. 2, pp. 1236–1248, Apr. 2023, doi: 10.1109/TAFFC.2021.3122146.
- [29] Y. Huang *et al.*, "FERMixNet: An Occlusion Robust Facial Expression Recognition Model with Facial Mixing Augmentation and Mid-Level Representation Learning," *IEEE Trans. Affect. Comput.*, 2024, doi: 10.1109/TAFFC.2024.3454102.
- [30] Z. Fei, B. Zhang, W. Zhou, X. Li, Y. Zhang, and M. Fei, "Global multi-scale extraction and local mixed multi-head attention for facial expression recognition in the wild," *Neurocomputing*, 2025, doi: 10.1016/j.neucom.2024.129323.
- [31] H. Tao and Q. Duan, "Hierarchical attention network with progressive feature fusion for facial expression recognition," *Neural Networks*, vol. 170, pp. 337–348, 2024, doi: 10.1016/j.neunet.2023.11.033.
- [32] D. Gera and S. Balasubramanian, "Landmark guidance independent spatio-channel attention and complementary context information based facial expression recognition," *Pattern Recognit. Lett.*, vol. 145, pp. 58–66, 2021, doi: 10.1016/j.patrec.2021.01.029.
- [33] Z. Zhao, Q. Liu, and S. Wang, "Learning Deep Global Multi-Scale and Local Attention Features for Facial Expression Recognition in the Wild," *IEEE Trans. Image Process.*, vol. 30, pp. 6544–6556, 2021, doi: 10.1109/TIP.2021.3093397.
- [34] Y. Tan, H. Xia, and S. Song, "Learning informative and discriminative semantic features for robust facial expression recognition," *J. Vis. Commun. Image Represent.*, vol. 98, 2024, doi: 10.1016/j.jvcir.2024.104062.
- [35] C. Wang, J. Xue, K. Lu, and Y. Yan, "Light Attention Embedding for Facial Expression Recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 4, pp. 1834–1847, 2022, doi: 10.1109/TCSVT.2021.3083326.
- [36] Y. Chen, K. Li, F. Tian, G. Wei, and M. Seberi, "Lightweight expression recognition combined attention fusion network with hybrid knowledge distillation for occluded e-learner facial images," *Neurocomputing*, vol. 628, 2025, doi: 10.1016/j.neucom.2025.129656.
- [37] Y. Wang *et al.*, "MGR3Net: Multigranularity Region Relation Representation Network for Facial Expression Recognition in Affective Robots," *IEEE Trans. Ind. Informatics*, vol. 20, no. 5, pp. 7216–7226, May 2024, doi: 10.1109/TII.2024.3353912.
- [38] D. Chen, G. Wen, H. Li, R. Chen, and C. Li, "Multi-Relations Aware Network for In-the-Wild Facial Expression Recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 8, pp. 3848–3859, 2023, doi: 10.1109/TCSVT.2023.3234312.
- [39] Q. Mao, L. Zhou, W. Zheng, X. Shao, and X. Huang, "Objective Class-Based Micro-Expression Recognition Under Partial Occlusion Via Region-Inspired Relation Reasoning Network," *IEEE Trans. Affect. Comput.*, vol. 13, no. 4, pp. 1998–2016, 2022, doi: 10.1109/TAFFC.2022.3197785.
- [40] Y. F. Huang and C. H. Tsai, "PIDViT: Pose-Invariant Distilled Vision Transformer for Facial Expression Recognition in the Wild," *IEEE Trans. Affect. Comput.*, vol. 14, no. 4, pp. 3281–3293, 2023, doi: 10.1109/TAFFC.2022.3220972.
- [41] P. Liu *et al.*, "Point Adversarial Self-Mining: A Simple Method for Facial Expression Recognition," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 12649–12660, 2022, doi: 10.1109/TCYB.2021.3085744.
- [42] Y. Zhao *et al.*, "Pose-invariant and occlusion-robust neonatal facial pain assessment," *Comput. Biol. Med.*, vol. 165, 2023, doi: 10.1016/j.combiomed.2023.107462.
- [43] K. Hu, G. Huang, Y. Yang, C. M. Pun, W. K. Ling, and L. Cheng, "Rapid facial expression recognition under part occlusion based on symmetric SURF and heterogeneous soft partition network," *Multimed. Tools Appl.*, vol. 79, no. 41–42, pp. 30861–30881, 2020, doi: 10.1007/s11042-020-09566-2.
- [44] L. Lyu *et al.*, "Spontaneous facial expression database of learners' academic emotions in online learning with hand occlusion," *Comput. Electr. Eng.*, vol. 97, 2022, doi: 10.1016/j.compeleceng.2021.107667.
- [45] A. R. Shahid and H. Yan, "SqueezeExpNet: Dual-stage convolutional neural network for accurate facial expression recognition with attention mechanism," *Knowledge-Based Syst.*, vol. 269, 2023, doi: 10.1016/j.knosys.2023.110451.
- [46] F. Ma, B. Sun, and S. Li, "Transformer-Augmented Network With Online Label Correction for Facial Expression Recognition," *IEEE Trans. Affect. Comput.*, vol. 15, no. 2, pp. 593–605, Apr. 2024, doi: 10.1109/TAFFC.2023.3285231.
- [47] W. Xie, H. Wu, Y. Tian, M. Bai, and L. Shen, "Triplet Loss with Multistage Outlier Suppression and Class-Pair Margins for Facial Expression Recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 2, pp. 690–703, Feb. 2022, doi: 10.1109/TCSVT.2021.3063052.
- [48] X. Yang, M. Han, Y. Luo, H. Hu, and Y. Wen, "Two-Stream Prototype Learning Network for Few-Shot Face Recognition Under Occlusions," *IEEE Trans. Multimed.*, vol. 25, pp. 1555–1563, 2023, doi: 10.1109/TMM.2023.3253054.
- [49] H. Li, N. Wang, X. Yang, X. Wang, and X. Gao, "Unconstrained Facial Expression Recognition With No-Reference De-Elements Learning," *IEEE Trans. Affect. Comput.*, vol. 15, no. 1, pp. 173–185, 2024, doi: 10.1109/TAFFC.2023.3263886.
- [50] F. Xue, Q. Wang, Z. Tan, Z. Ma, and G. Guo, "Vision Transformer With Attentive Pooling for Robust Facial Expression Recognition," *IEEE Trans. Affect. Comput.*, vol. 14, no. 4, pp. 3244–3256, 2023, doi: 10.1109/TAFFC.2022.3226473.
- [51] A. B. Ahadit and R. K. Jatoth, "A novel dual CNN architecture with LogicMax for facial expression recognition," *J. Inf. Sci. Eng.*, vol. 37, no. 1, pp. 15–39, 2021, doi: 10.6688/JISE.202101_37(1).0002.
- [52] L. Xiong, J. Zhang, X. Zheng, and Y. Wang, "Context Transformer and Adaptive Method with Visual Transformer for Robust Facial Expression Recognition," *Appl. Sci.*, vol. 14, no. 4, 2024, doi: 10.3390/app14041535.
- [53] Y. Cheng and D. Kong, "CSINet: Channel-Spatial Fusion Networks for Asymmetric Facial Expression Recognition," *Symmetry (Basel)*, vol. 16, no. 4, p. 471, Apr. 2024, doi: 10.3390/sym16040471.
- [54] C. Liu, X. Liu, C. Chen, and K. Zhou, "Deep Global Multiple-Scale and Local Patches Attention Dual-Branch Network for Pose-Invariant Facial Expression Recognition," *C. - Comput. Model. Eng. Sci.*, vol. 139, no. 1, pp. 405–440, 2023, doi: 10.32604/cmes.2023.031040.
- [55] S. B. Sukhvasi, S. B. Sukhvasi, K. Elleithy, A. El-Sayed, and A. Elleithy, "Deep Neural Network Approach for Pose, Illumination, and Occlusion Invariant Driver Emotion Detection," *Int. J. Environ. Res. Public Health*, vol. 19, no. 4, 2022, doi: 10.3390/ijerph19042352.
- [56] Z. Ullah *et al.*, "Emotion Recognition from Occluded Facial Images Using Deep Ensemble Model," *Comput. Mater. Contin.*, vol. 73, no. 3, pp. 4465–4487, 2022, doi: 10.32604/cmc.2022.029101.
- [57] Y. Tian, J. Zhu, H. Yao, and D. Chen, "Facial Expression Recognition Based on Vision Transformer with Hybrid Local Attention," *Appl. Sci.*, vol. 14, no. 15, 2024, doi: 10.3390/app14156471.
- [58] L. Ruan, Y. Han, J. Sun, Q. Chen, and J. Li, "Facial expression recognition in facial occlusion scenarios: A path selection multi-network," *Displays*, vol. 74, 2022, doi: 10.1016/j.displa.2022.102245.

- [59] S. Bellamkonda, N. P. Gopalan, C. Mala, and L. Settipalli, "Facial expression recognition on partially occluded faces using component based ensemble stacked CNN," *Cogn. Neurodyn.*, vol. 17, no. 4, pp. 985–1008, Aug. 2023, doi: 10.1007/s11571-022-09879-y.
- [60] X. Li, C. Zhu, and F. Zhou, "Facial Expression Recognition: One Attention-Modulated Contextual Spatial Information Network," *Entropy*, vol. 24, no. 7, p. 882, Jun. 2022, doi: 10.3390/e24070882.
- [61] K. Y. Tsai, Y. W. Tsai, Y. C. Lee, J. J. Ding, and R. Y. Chang, "Frontalization and adaptive exponential ensemble rule for deep-learning-based facial expression recognition system," *Signal Process. Image Commun.*, vol. 96, no. April, p. 116321, 2021, doi: 10.1016/j.image.2021.116321.
- [62] T. Cao, C. Liu, and J. Chen, "Nonfrontal Expression Recognition in the Wild Based on PRNet Frontalization and Muscle Feature Strengthening," *Math. Probl. Eng.*, vol. 2021, pp. 1–21, Jul. 2021, doi: 10.1155/2021/6620752.
- [63] S. S. Sudha and S. S. Suganya, "On-road driver facial expression emotion recognition with parallel multi-verse optimizer (PMVO) and optical flow reconstruction for partial occlusion in internet of things (IoT)," *Meas. Sensors*, vol. 26, 2023, doi: 10.1016/j.measen.2023.100711.
- [64] C. Ge, "Overcoming occlusions in complex environments to achieve robust perception of human emotions," *Eng. Res. Express*, vol. 6, no. 4, 2024, doi: 10.1088/2631-8695/ad9fd6.
- [65] J. Gao and Y. Zhao, "TFE: A Transformer Architecture for Occlusion Aware Facial Expression Recognition," *Front. Neurobot.*, vol. 15, 2021, doi: 10.3389/fnbot.2021.763100.
- [66] K. Zheng, L. Tian, Z. Li, H. Li, and J. Zhang, "Incorporating eyebrow and eye state information for facial expression recognition in mask-obscured scenes," *Electron. Res. Arch.*, vol. 32, no. 4, pp. 2745–2771, 2024, doi: 10.3934/ERA.2024124.
- [67] S. Li, W. Deng, and J. P. Du, "Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild," *Proc. - 30th IEEE Conf. Comput. Vis. Pattern Recognition, CVPR 2017*, vol. 2017-Janua, pp. 2584–2593, 2017, doi: 10.1109/CVPR.2017.277.
- [68] P. Lucey, J. F. Cohn, T. Kanade, J. Saragih, Z. Ambadar, and I. Matthews, "The extended Cohn-Kanade dataset (CK+): A complete dataset for action unit and emotion-specified expression," *2010 IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. - Work. CVPRW 2010*, no. July, pp. 94–101, 2010, doi: 10.1109/CVPRW.2010.5543262.
- [69] A. Mollahosseini, B. Hasani, and M. H. Mahoor, "AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild," *IEEE Trans. Affect. Comput.*, vol. 10, no. 1, pp. 18–31, 2019, doi: 10.1109/TAFFC.2017.2740923.
- [70] E. Barsoum, C. Zhang, C. C. Ferrer, and Z. Zhang, "Training deep networks for facial expression recognition with crowd-sourced label distribution," *ICMI 2016 - Proc. 18th ACM Int. Conf. Multimodal Interact.*, pp. 279–283, 2016, doi: 10.1145/2993148.2993165.
- [71] I. J. Goodfellow et al., "Challenges in representation learning: A report on three machine learning contests," *Neural Networks*, vol. 64, pp. 59–63, 2015, doi: 10.1016/j.neunet.2014.09.005.
- [72] M. Kamachi, M. Lyons, and J. Gyoba, "The Japanese female facial expression (jaffe) database," *Proc. 3rd Int. Conference Autom. Face Gesture Recognit.*, vol. 21, no. January, p. 32, 1998, [Online]. Available: <https://zenodo.org/records/3451524>
- [73] A. Dhali, R. Goecke, S. Lucey, and T. Gedeon, "Static facial expression analysis in tough conditions: Data, evaluation protocol and benchmark," *Proc. IEEE Int. Conf. Comput. Vis.*, pp. 2106–2112, 2011, doi: 10.1109/ICCVW.2011.6130508.
- [74] G. Zhao, X. Huang, M. Taini, S. Z. Li, and M. Pietikäinen, "Facial expression recognition from near-infrared videos," *Image Vis. Comput.*, vol. 29, no. 9, pp. 607–619, 2011, doi: 10.1016/j.imavis.2011.07.002.
- [75] A. K. Davison, C. Lansley, N. Costen, K. Tan, and M. H. Yap, "SAMM: A Spontaneous Micro-Facial Movement Dataset," *IEEE Trans. Affect. Comput.*, vol. 9, no. 1, pp. 116–129, 2018, doi: 10.1109/TAFFC.2016.2573832.
- [76] G. B. Huang, M. Mattar, T. Berg, and E. Learned-Miller, "Labeled Faces in the Wild: A Database for Studying Face Recognition in Unconstrained Environments," in *Workshop on Faces in "Real-Life" Images: Detection, Alignment, and Recognition*, Marseille, France, Oct. 2008. [Online]. Available: <https://inria.hal.science/inria-00321923>
- [77] O. Langner, R. Dotsch, G. Bijlstra, D. H. J. Wigboldus, S. T. Hawk, and A. van Knippenberg, "Presentation and validation of the Radboud Faces Database," *Cogn. Emot.*, vol. 24, no. 8, pp. 1377–1388, 2010, doi: 10.1080/02699930903485076.
- [78] V. Jain, J. Crowley, and others, "Smile detection using multi-scale gaussian derivatives," *12th WSEAS Int. Conf. Signal Process. Robot. Autom.*, pp. 149–154, 2013.
- [79] Z. Wang, B. Huang, G. Wang, P. Yi, and K. Jiang, "Masked Face Recognition Dataset and Application," *IEEE Trans. Biometrics, Behav. Identity Sci.*, vol. 5, no. 2, pp. 298–304, 2023, doi: 10.1109/TBIOM.2023.3242085.
- [80] X. Jiang et al., "DFEW: A Large-Scale Database for Recognizing Dynamic Facial Expressions in the Wild," in *MM 2020 - Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 2881–2889. doi: 10.1145/3394171.3413620.
- [81] Z. Liu, P. Luo, X. Wang, and X. Tang, "Deep learning face attributes in the wild," *Proc. IEEE Int. Conf. Comput. Vis.*, vol. 2015 Inter, pp. 3730–3738, 2015, doi: 10.1109/ICCV.2015.425.
- [82] D. Yi, Z. Lei, S. Liao, and S. Z. Li, "Learning Face Representation from Scratch," 2014, [Online]. Available: <http://arxiv.org/abs/1411.7923>
- [83] S. Du, Y. Tao, and A. M. Martinez, "Compound facial expressions of emotion," *Proc. Natl. Acad. Sci. U. S. A.*, vol. 111, no. 15, 2014, doi: 10.1073/pnas.1322355111.
- [84] W. J. Yan et al., "CASME II: An improved spontaneous micro-expression database and the baseline evaluation," *PLoS One*, vol. 9, no. 1, pp. 1–8, 2014, doi: 10.1371/journal.pone.0086041.
- [85] M. Pantic, M. Valstar, R. Rademaker, and L. Maat, "Web-based database for facial expression analysis," *IEEE Int. Conf. Multimed. Expo, ICME 2005*, vol. 2005, pp. 317–321, 2005, doi: 10.1109/ICME.2005.1521424.
- [86] D. Kollias and S. Zafeiriou, "Analysing Affective Behavior in the second ABAW2 Competition," *Proc. IEEE Int. Conf. Comput. Vis.*, vol. 2021-Octob, pp. 3645–3653, 2021, doi: 10.1109/ICCVW54120.2021.00408.
- [87] R. Beekman, C. C. Tijssen, L. H. Visser, and R. L. L. A. Schellens, "Dropped head as the presenting symptom of primary hyperparathyroidism [3]," *J. Neurol.*, vol. 249, no. 12, pp. 1738–1739, 2002, doi: 10.1007/s00415-002-0898-7.
- [88] T. Cao, C. Liu, J. Chen, and L. Gao, "Nonfrontal and Asymmetrical Facial Expression Recognition through Half-Face Frontalization and Pyramid Fourier Frequency Conversion," *IEEE Access*, vol. 9, pp. 17127–17138, 2021, doi: 10.1109/ACCESS.2021.3052500.



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Half-Flip 2D for Robust Facial Emotion Recognition in Engagement Evaluation during Online Learning

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ABSTRACT Engagement evaluation in online learning relies heavily on facial emotion recognition (FER) to capture learner attentiveness, affective involvement, and cognitive focus. However, FER performance is often disrupted when students partially cover their faces with hands or adopt non-frontal poses, conditions frequently observed during instructional sessions. Such occlusions and pose variations reduce recognition reliability and weaken engagement assessment. To address this challenge, this study introduces the Half-Flip 2D technique, which duplicates the visible half of an occluded face in 2D and merges it into a synthetic frontal image, enabling recovery of key expression features even under vertical occlusion or rotation. The method is evaluated using CNN, DeepFace, FaceLib, and SVM on datasets representing typical online learning scenarios. Experimental results show substantial accuracy improvements in vertical occlusion: 33.79% for CNN, 14.29% for DeepFace, 10.66% for FaceLib, and 6.49% for SVM. For non-frontal poses, improvements reached 16.53% for CNN and 2.21% for SVM. By reducing recognition errors, Half-Flip 2D enhances the stability of emotion-based engagement metrics, thereby supporting more consistent analytics in asynchronous learning environments and enabling adaptive instructional strategies through reliable engagement detection.

INDEX TERMS Engagement, facial emotion recognition, half-flip 2d, occlusion, online learning.

I. INTRODUCTION

Facial Emotions Recognition (FER) has emerged as a significant area of interest within the educational domain, particularly in the context of asynchronous online learning. Asynchronous online learning refers to instructional activities that are not bound by specific time constraints, enabling learners to access and engage with materials at any time and from any location according to their availability [1]. Within this learning mode, FER serves to identify and interpret facial expressions such as happiness, sadness, or confusion in order to evaluate levels of student engagement. In asynchronous environments, FER facilitates the automatic monitoring of learners' emotional states without requiring direct instructor intervention, thereby supporting the provision of more personalized and adaptive feedback.

FER has demonstrated its capability to accurately identify and interpret facial emotions [2]. Nevertheless, several challenges remain, particularly in ensuring that FER systems

can maintain reliable performance when facial features are partially obscured by external factors such as hands, hair, head tilting, or when learners are not directly facing the camera. Such conditions may occur as a result of student disengagement, distractions, or health-related issues that cause discomfort in the neck and head [3]. Under these circumstances, the precision of emotion detection is likely to decline significantly, thereby reducing the overall effectiveness of FER in educational applications.

The problem formulation in this study focuses on addressing the limitations that continue to hinder the effectiveness of Facial Emotions Recognition (FER) in educational contexts. Although FER has proven capable of identifying and interpreting facial emotions, its performance decreases significantly under non-ideal conditions. Specifically, challenges arise when a learner's face is partially occluded by objects (such as hands or hair), tilted, or not directly oriented forward the camera (FIGURE 1).

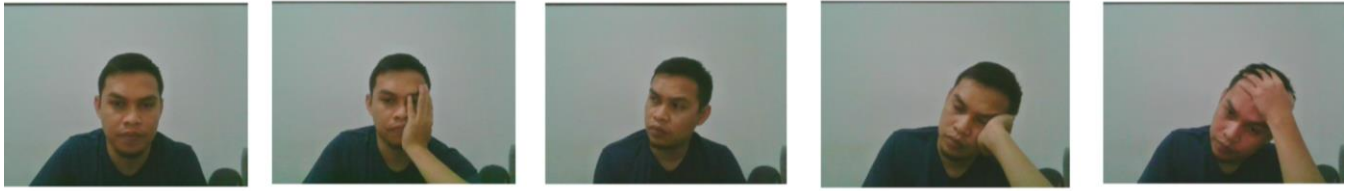


FIGURE 1. Conditions of hand and head during online learning

These circumstances are common in online learning environments, where students may experience boredom, distractions, or physical discomfort that affects their posture and facial visibility. Consequently, the accuracy of FER in detecting and interpreting emotions is compromised, reducing its reliability as a tool for monitoring engagement and providing adaptive feedback. Therefore, the central problem addressed in this study is how to develop or refine FER systems so that they remain effective in detecting and interpreting facial emotions even when facial visibility is limited or distorted (occluded face).

A variety of strategies have been introduced to mitigate the challenges posed by occlusion in FER. Among these are deep learning (DL) models trained with occlusion-based augmentation to enhance robustness [4], as well as multimodal approaches that integrate facial imagery with complementary data sources such as audio signals [5][6]. While these techniques have demonstrated considerable promise, they frequently rely on computationally intensive processes that can increase latency during FER operations. This underscores the ongoing need for more streamlined yet effective solutions capable of addressing occlusion without incurring substantial computational costs.

An additional component of the FER system pipeline that can be enhanced is the pre-processing stage, particularly through facial reconstruction prior to recognition. Existing studies suggest that reconstruction can be achieved using Generative Adversarial Networks (GANs), which operate by identifying occluding objects, removing them, and subsequently filling the missing regions with plausible facial features derived from previously trained data [7]. While this technique has the potential to improve recognition accuracy, it necessitates training on datasets composed of unobstructed facial images. This requirement limits the practicality of GAN-based reconstruction, especially in resource-constrained environments.

Another potential enhancement in the pre-processing stage of FER systems is facial frontalization, which can be implemented through either 3D reconstruction techniques [8] or 2D transformations [9]. The purpose of frontalization is to reorient a laterally positioned face so that it appears forward-facing. While this method has been shown to mitigate the decline in recognition accuracy associated with non-frontal facial orientations [10], it remains insufficient for addressing occlusion-related challenges. In particular, frontalization cannot effectively handle cases where facial regions are obstructed by external objects, such as hands partially covering the face during online learning sessions.

Based on the findings of Sukumaran et al. (2024), engagement level classification through a multimodal approach that utilizes facial expression data, head orientation, and eye condition has proven effective in achieving accurate detection [11]. This study adopts similar multimodal parameters to those employed in the aforementioned research, but introduces an additional pre-processing stage prior to facial emotion detection. This stage involves the removal of hand occlusion objects and the application of facial frontalization techniques. As a result, the accuracy of FER remains high even when faces are occluded or not in a frontal position, thereby enhancing the overall performance of the engagement detection system. The primary contribution of this study lies in the introduction of a novel pre-processing strategy that integrates frontalization and hand-object removal using 2D image processing prior to emotion classification by the FER model, thereby ensuring robust facial feature extraction and ultimately improving the overall performance of multimodal engagement detection.

II. RELATED WORKS

A. ENGAGEMENT DETECTION IN ONLINE LEARNING

Research on student engagement detection in online learning has increasingly focused on multimodal approaches that utilize diverse data sources to improve system accuracy and reliability. Such approaches extend beyond facial expression analysis, incorporating behavioral, audio, and physiological signals [12]. Evidence suggests that combining facial expression analysis, eye-blink frequency, and head movement enables real-time engagement detection in e-learning sessions. These methods employ deep convolutional neural networks (CNNs) and landmark detection to process webcam data, producing a more accurate engagement index and facilitating immediate feedback for learners [13].

Recent developments include multimodal engagement recognition systems based on webcam imagery that integrate facial expression detection, gaze tracking, head pose estimation, and blink rate analysis through deep learning techniques. Designed specifically for e-learning environments, these systems have demonstrated significant improvements in engagement recognition compared to traditional unimodal models [11]. Overall, the integration of multimodal inputs like visual, audio, and behavioral emerges as a critical strategy for automatically and adaptively detecting and monitoring student engagement in digital learning contexts, thereby establishing a strong foundation for more responsive and personalized online education.

B. FORMS OF OCCLUSION IN FER SYSTEMS

Forms of occlusion in facial emotion recognition (FER) systems include head pose variation [14], illumination variation [15], medical masks [16], hair and glasses [17], [18], or other facial accessories. All these forms of occlusion have the potential to reduce the accuracy of the FER system. Occlusion objects or conditions can either be obtained directly from the dataset (natural) or synthetically created.

Natural occlusion refers to conditions that occur naturally in the real-world environment and are generally uncontrollable by the system, such as head pose variation, lighting variation, and other attributes like hair and glasses. Studies [14], [15], [19], [20], [21], [22], and [8] have shown that changes in head position can significantly block key areas of the face, thus reducing the accuracy of the FER system. Similar findings were also noted in lighting variation [15], which can lead to inconsistency in visual features, as well as personal attributes such as hair and glasses [17], [15], which obscure key areas of the face like the eyes and eyebrows.

Synthetic occlusion refers to conditions that are intentionally created and simulated in experiments to assess the robustness of the FER system under extreme conditions. Common examples include medical masks [16], [23], [24], which cover the mouth and chin areas, masking blocks [25], [26], involving the creation of black block objects to randomly cover areas of the face, and occlusion in specific regions such as the eyes and hair [27], [28]. This type of occlusion is designed to evaluate the impact of covering specific facial areas on FER accuracy, helping to analyze the system's sensitivity to the loss of visual information from the face.

C. APPROACHES IN OCCLUSION FOR FER SYSTEMS

In recent years, research on how to address occlusion issues in FER systems has been dominated by the creation of models that are robust to such occlusions. Models are trained with diverse face data (including occlusion conditions) using public datasets that contain various types of occlusions, such as faces covered by objects, faces turned sideways, or faces under extreme lighting conditions. Some approaches undertaken by researchers include using local attention features [20][29][30], augmenting training data by mixing specific parts of the face [31], or combining several deep learning architectures to build the model, such as CNN+ViT [32] or CNN+LSTM [23].

Another approach researchers have used to overcome occlusion is in the pre-processing stage of the FER system. For instance, Cao et al. took half of the face and combined it in a 3D form [8], followed by muscle enhancement around the nose area to improve emotion recognition accuracy from 3D face data. The results showed improved accuracy for some basic emotions after the muscle enhancement process around the nose. Other 3D processing was also conducted by Ning Sun et al. They performed face frontalization using 3D face landmarks [14], and in general, the accuracy results showed improvements compared to other state-of-the-art methods. However, 3D image processing has limitations in

terms of resource requirements, as it demands higher resources due to more complex computations compared to 2D image processing [9].

Face reconstruction in 2D has also been conducted by some researchers, such as Delphine Poux et al. [28]. They created a synthetic occlusion that covered part of the face with a black object. The method they proposed used optical flow reconstruction, which is a flow that can estimate and fill the occlusion shape, thereby creating a face image with complete features, including the eyes, nose, and mouth. Although the results were not stated to be better than other state-of-the-art methods, the proposed method is competitive in removing occlusions.

Another approach in 2D involves using two flipped (horizontal flip) face images and combining them to appear as a frontal face [22]. This research was conducted by Kai-Yuan Tsai et al. in 2021. The results of the frontalization appeared quite smooth, but the solution they developed was not tested with occlusion types such as hands or other forms, so its reliability in occlusion conditions has not been validated.

D. DATASETS USED IN DEVELOPING FER SYSTEMS WITH OCCLUSION

A wide range of datasets has been employed to study occlusion in facial expression recognition (FER), RAF-DB [33], CK+ [34], AffectNet-7 [35], FERPlus [36], and FER2013 [37] among the most widely used. These datasets provide large collections of facial images annotated with emotion labels, supporting the development and evaluation of robust FER models. CK+, for instance, contains 593 facial expression sequences from 123 individuals, enabling dynamic analysis of expressions from neutral to peak emotions. AffectNet-7, with over 450,000 annotated images, offers one of the most comprehensive resources for emotion recognition, while FER2013 and its enhanced version FERPlus provide benchmark datasets of 48×48 pixel grayscale images, with FERPlus offering refined labels across eight emotion categories.

Beyond general datasets, specialized corpora have been developed to evaluate FER robustness under occlusion and pose variations. Wang et al. (2020) [38] introduced Occlusion-AffectNet, Occlusion-RAF-DB, Occlusion-FERPlus, Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus, specifically designed to simulate real-world challenges such as partial face coverings and non-frontal orientations. These datasets allow researchers to systematically assess how models perform when facial features are obstructed or when faces deviate from frontal alignment.

Occlusion-based datasets introduce artificial or natural obstructions, such as hands, masks, or glasses, to test recognition accuracy when critical facial landmarks are hidden. For example, Occlusion-FERPlus targets occlusions of the eyes, nose, or mouth, while Occlusion-RAF-DB incorporates both synthetic and real-world occlusions. Similarly, Occlusion-AffectNet evaluates model resilience against diverse occlusion scenarios, reflecting practical

conditions encountered in online learning or everyday interactions.

Pose-focused datasets, including Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus, emphasize variations in head orientation, such as tilted or sideways faces. These resources are crucial for analyzing how pose affects emotion classification accuracy and for developing models capable of generalizing across diverse viewing angles. Collectively, occlusion and pose datasets provide essential benchmarks for advancing FER systems toward greater robustness and applicability in real-world contexts.

III. MATERIALS AND METHOD

A. DATASET

In this work, we utilize the AffectNet and RAF-DB datasets for testing purposes. However, instead of using all the available data from these datasets, we focus on selecting only the relevant data by applying the Py-Feat tool [39] to clean and filter the datasets according to their labels. This ensures that the datasets used for testing contain only well-labeled and accurate facial expressions, thus improving the quality of our evaluation. By carefully curating the data, we aim to minimize any noise or inconsistencies that could affect the results and the reliability of the experiments.

Additionally, to address the varying face orientations in the datasets, we separate frontal faces from side-facing faces using the mediapipe library [40]. This library enables estimation of facial gaze direction, facilitating the categorization of images into frontal and non-frontal groups. This distinction allows for a more thorough evaluation of the proposed method across various real-world conditions, such as side-turned or partially occluded faces. Such categorization is essential for assessing the model's generalization capability in practical contexts, including online learning and video conferencing.

B. PROPOSED METHOD

We propose a pre-processing method called **Half-Face2D**, which is a face frontalization technique that also removes hand occlusions partially covering the face. **FIGURE 2** illustrates the process, starting from an input image (a) to the final output (l). The first step is to perform head roll correction to generate an upright head image (b). The next step is to perform landmarking on image (b), resulting in face landmarks (c) and hand landmarks (d). Then, hand blurring is applied to image (b) based on hand landmarks (d), producing image (e), which will be used as the base image for the final processing stage.

The next step is to mask the face and hand (f), with the face masked in white and the hand masked in red. The resulting masked image will be divided into two parts based on the central points of the face according to the face landmarks (c). The part with the greater number of white pixels will be selected and its middle section will be stretched. Initially, the middle part of the face follows the facial contour, but it will be made upright (with the same X

coordinates). The stretched image will be overlaid with the original face image (c), resulting in image (i). Image (i) is then duplicated and merged to form the complete, unoccluded, and upright face image (l).

Algorithm 1 Half-Flip2D Image Processing

Input:

I_{in} (hand-occluded face or face with side view image)

Step 1: //roll head correction

$I_{rot} \leftarrow \text{RotateForRollCorrection}(I_{in})$

Step 2: //face and hand landmarking

$L_{face}, L_{hand} \leftarrow \text{DetectLandmarks}(I_{rot})$

$I_{src} \leftarrow I_{rot}$

Step 3: //face and hand masking, white for face, red for hand

$M_{face} \leftarrow \text{CreateFaceMask}(I_{rot}, L_{face}, \text{WHITE})$

if $L_{hand} = \text{TRUE}$ then

$M_{hand} \leftarrow \text{CreateHandMask}(I_{rot}, L_{hand}, \text{RED})$

$I_{src} \leftarrow \text{ApplyHandBlurring}(I_{rot}, L_{hand})$

$M_{full} \leftarrow \text{CombineMasks}(M_{face}, M_{hand})$

else

$M_{full} \leftarrow M_{face}$

end if

Step 4: //devide face to two region

$M_{left}, M_{right} \leftarrow \text{DivideByCenter}(M_{full}, L_{face})$

$CM_{left} \leftarrow \text{CountWhitePixels}(M_{left})$

$CM_{right} \leftarrow \text{CountWhitePixels}(M_{right})$

Step 5: //compare two region of the face

if $CM_{left} > CM_{right}$ then

$M_{sel} \leftarrow M_{left}$

Else

$M_{sel} \leftarrow M_{right}$

end if

Step 6: //stretch the center poin of the selected region face

$M_{stretch} \leftarrow \text{StretchToStraightLine}(M_{sel})$

Step 7: //replace masked face with the original face source

$I_{sel} \leftarrow \text{ReplaceRegion}(I_{src}, M_{stretch})$

Step 8: //duplicate and flip the image

$I_{flip} \leftarrow \text{DuplicateFlipHorizontal}(I_{sel})$

Step 9: //combine output of step 7 and step 8, then resize

$I_{full} \leftarrow \text{Combine}(I_{sel}, I_{flip})$

$I_{out} \leftarrow \text{ResizeImage}(I_{full}, 224, 224)$

Output: I_{out} (non-occluded and frontal face)

The flow of the proposed method is further detailed in Algorithm 1. The “DetectLandmark” function in Algorithm 1 serves as a substitute for traditional face detection techniques such as MTCNN, RetinaFace, or Haarcascade models. The underlying logic is that if facial landmarks are not detected, the image will be classified as not containing a face. This function plays a critical role in ensuring the accuracy and efficiency of the proposed method. For a clearer understanding of the process, the visualization of the proposed method can be observed in **FIGURE 2**.

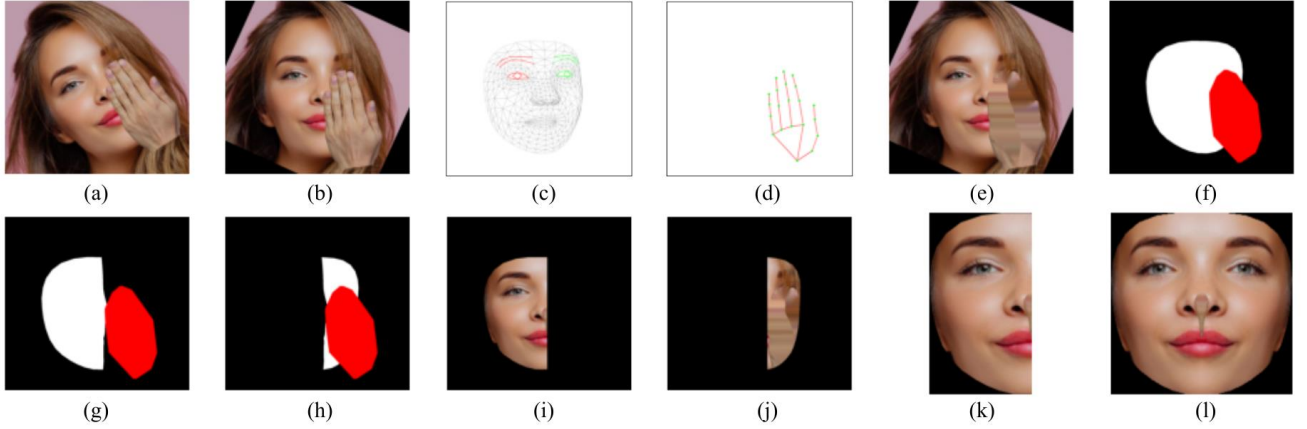


FIGURE 2. Steps for processing of proposed method, start from input image (a) and finish with the output (l)

IV. METHODOLOGY

A. DATA SELECTION

The first step in our methodology involves face selection from the datasets. We began by combining the AffectNet and RAF-DB datasets to create a comprehensive data pool for testing. However, given the presence of inconsistencies and mislabeled data in the AffectNet dataset, it was necessary to apply a cleaning procedure. To address this, we utilized the Py-Feat tool [39][39], which facilitated the filtering and selection process, ensuring that only the correctly labeled and relevant facial data were retained for further analysis. This cleaning process is crucial, as it removes any erroneous or misclassified samples, which could otherwise compromise the integrity of the model's performance and the validity of our results. The outcome of this cleaning process is illustrated in FIGURE 3, which demonstrates the refinement of the dataset to ensure accurate and reliable testing.



FIGURE 3. Inaccurate facial annotations in the AffectNet dataset

After that, we separated the dataset into frontal face data and non-frontal face data. The separation was done by calculating the yaw angle of the head using the mediapipe library. If the yaw angle is less than 30 degrees, it is considered a frontal face, otherwise it is considered a non-frontal face. FIGURE 4 shows the results of the selection process.



FIGURE 4. (a) Frontal face (b) Non-frontal face

B. HAND SYNTHESIS

To evaluate the effectiveness of the proposed method in removing hand occlusions and performing frontalization, this study requires a dataset with vertical hand occlusion covering half of the face. We conducted augmentation by randomly adding synthetic hands to the frontal face images as shown in FIGURE 5.



FIGURE 5. (a) Raw image (b) Hand synthetic image (c) Output image

This study involves two types of data for testing, face with partial hand occlusion and non-frontal face, as shown in FIGURE 6. The dataset includes 1,400 images of faces with partial hand occlusion and 1,043 images of side-facing faces, with the distribution of each emotion detailed in TABLE I.

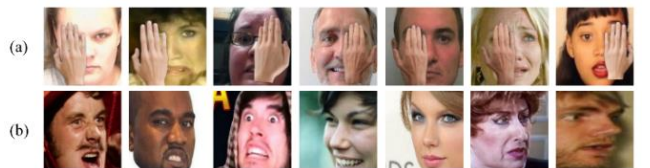


FIGURE 6. Testing data (a) Faces with hand occlusion (b) Faces with side view

TABLE I
TEST DATA DISTRIBUTION

Fig. 6	An	Di	Fe	Ha	Ne	Sa	Su
(a)	200	200	200	200	200	200	200
(b)	72	138	98	200	200	157	178

C. PROPOSED METHOD IMPLEMENTATION

The next step is to implement the proposed method on the test data (FIGURE 6) as described in Algorithm 1 and illustrated in FIGURE 5. The testing was conducted on the entire prepared test dataset in RGB image format with various background conditions. The output remains in RGB format, but the background is removed and replaced with a black background (FIGURE 7), ensuring that the resulting image focuses solely on the face shape before classification with the FER model.

D. FER MODEL

We also developed a FER model with a Convolutional Neural Network (CNN) architecture using only frontal faces from the AffectNet dataset, with 350 images per emotion class (FIGURE 7). The goal was to test the impact of facial occlusion, specifically faces partially covered by hands and faces in non-frontal positions.



FIGURE 7. Training data

The dataset was split into 80% for training and 20% for validation. The model training was conducted using the early stopping technique, which aims to automatically stop the training process if the validation accuracy does not show improvement or even decreases. This method helps prevent overfitting by ensuring that the model does not train too long on the training data, which could lead to performance degradation on the validation data. The training process utilized the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and a maximum of 50 epochs, adjusted to ensure stable model convergence without sacrificing generalization.

E. EVALUATION

The evaluation was conducted by measuring the accuracy of facial emotion recognition by calculating the difference in accuracy between the data before and after the proposed method was applied. This approach was used to assess the impact of the proposed method, which may either improve or decrease the accuracy.

We used four FER models to test the proposed method. First, testing was conducted with the custom CNN model, which was trained exclusively on frontal face data, followed by testing with DeepFace [41], FaceLib [42], and SVM. The testing was conducted separately between the data with hand occlusion and the data with non-frontal faces. Specifically, for non-frontal faces, we also compared the results from our proposed method with the results from the 2D frontalization method performed by Kai-Yuan Tsai et al. (2021) [22]. The testing on the data with hand occlusion and non-frontal face was conducted using the combination of the RAF-DB and

AffectNet datasets. WAdditionally, we will conduct testing on data with horizontal hand occlusion (covering the lower part/mouth), a common occurrence in real-world scenarios, as shown in FIGURE 8.



FIGURE 8. Horizontal hand-occlusion face

F. ENGAGEMENT CALCULATION

We will integrate the results of the proposed method into the implementation of FER within an online learning system using emotion data, head orientation, and eye gaze [11]. The weighting (TABLE II – IV) is applied to determine the students' engagement status using an engagement index.

TABLE II
WEIGHT OF EMOTIONS

Emotions	Ang	Dis	Fea	Hap	Neu	Sad	Sur
Weight	0.1	0.9	0.5	1.1	1.4	0.3	0.7

TABLE III
WEIGHT OF GAZE DIRECTION

Look Center	Not Look Center
1	0

TABLE IV
WEIGHT OF EYE STATUS

Eye Open	Eye Close
2.5	0

The Egement Index (EI) will be calculated for each facial frame using Equation 1.

$$EI = g(i) + e(i) + f(i) \quad (1)$$

where EI is the Engagement Index value, $g(i)$ is the weight of the gaze direction, $e(i)$ is the weight of the eye status, and $f(i)$ is the weight of the facial emotion obtained from the FER model classification. Each frame will have its engagement status determined using the rules in TABLE V. The overall conclusion of a student's engagement will be determined by the most dominant engagement status. An overview of this entire work can be seen in FIGURE 9.

TABLE V
ENGAGEMENT STATUS CLASSIFICATION BASED ON EI VALUE

Engagement status	EI
Highly Engaged	$EI \geq 4.5$
Confused	$4 \leq EI < 4.5$
Boredom	$2.5 \leq EI < 4$
Sleepy	$0 < EI < 2.5$
Very Not Engaged	$EI = 0$

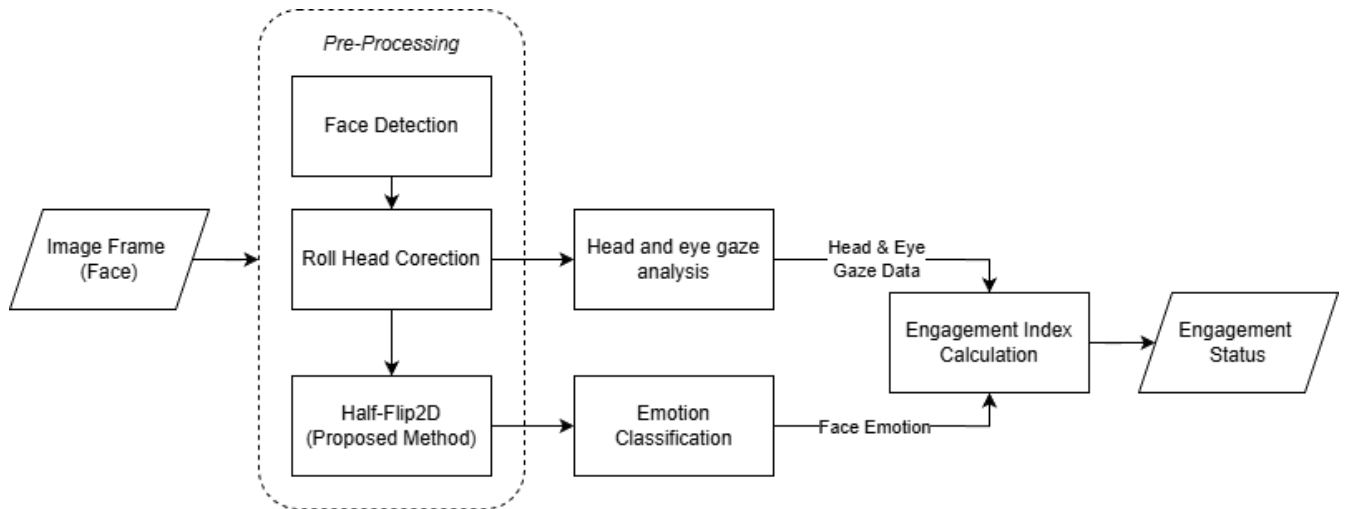


FIGURE 9. The workflow of the engagement assessment system in online learning

V. RESULT

A. CNN MODEL RESULT

Based on the experimental results of creating the FER model with a CNN architecture, the training was stopped after 26 iterations, with a training accuracy of 0.88 and a validation accuracy of 0.65, as shown in FIGURE 10. This result indicates that while the model achieved good accuracy on the training data, its ability to generalize to validation data still needs improvement.

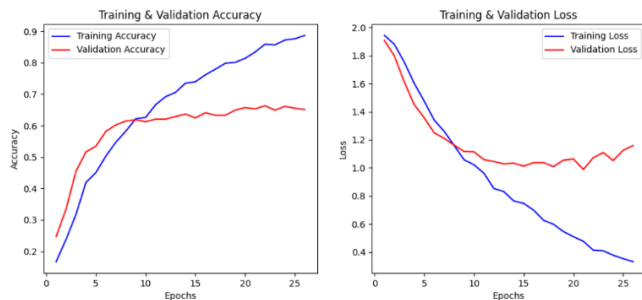


FIGURE 10. Train and validation accuracy result

This condition was deliberately created to test the model's weaknesses in validation with face data partially occluded by a vertical hand and non-frontal face conditions, thereby providing an alternative approach to address the limitations of an emotion recognition model in online learning through pre-processing techniques as proposed in our method.

Based on the results of the confusion matrix (FIGURE 11) obtained from the emotion expression recognition model trained using only frontal face data without augmentation, it can be concluded that the model shows fairly good accuracy on some dominant emotion classes, such as happy and angry. However, there is still overlap in classification among classes with similar visual features, such as fear, surprise, and neutral. This aligns with research conducted by Szczakowska et al. (2023) [43], which emphasizes that without data augmentation and diversification, models tend to be biased toward the majority class and struggle to distinguish emotions with similar

features, thus leading to suboptimal classification performance for minority classes. Therefore, the application of data augmentation strategies is recommended to improve generalization ability and reduce classification errors between classes, as highlighted in recent studies related to facial emotion recognition based on deep learning.

As an alternative to data augmentation, which can increase computational load and extend training time [44], this study adopts a pre-processing approach before the emotion recognition process by the model. By performing standardization, normalization, and other transformations on facial images prior to the classification stage, the pre-processing technique has been empirically shown to improve both the accuracy and efficiency of the emotion recognition system, as well as help reduce classification errors between classes [45].

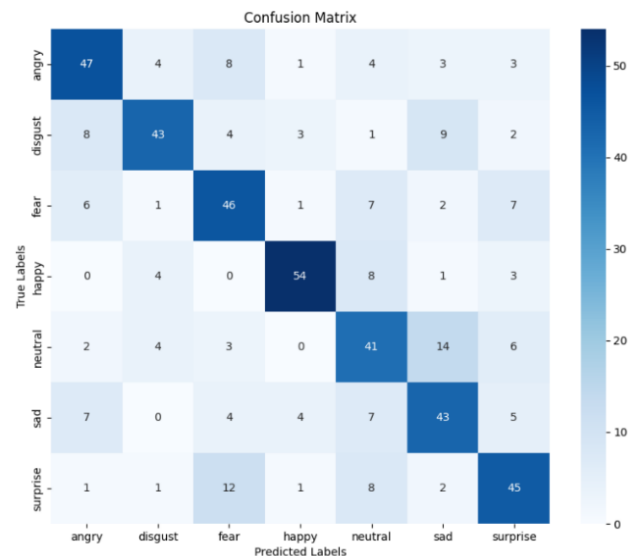


FIGURE 11. Confusion matrix of CNN model

B. PROPOSED METHOD RESULT

This section presents the results of the face pre-processing experiments conducted on images featuring hand occlusion

and non-frontal poses, as outlined in the proposed method. The pre-processing pipeline aims to address common challenges encountered in real-world scenarios, where faces may be partially covered or not oriented directly towards the camera. By simulating these challenging conditions, the evaluation provides insight into the effectiveness of the developed approach.

FIGURE 12 demonstrates the outcome of the pre-processing stage for an image exhibiting both hand occlusion and a non-frontal pose. The input image is systematically processed to mitigate occlusion effects and to align the facial region prior to classification by the FER model. Such pre-processing is crucial for enhancing the accuracy of emotion recognition, as it improves the model's ability to extract meaningful features despite the presence of occlusions or pose variations.

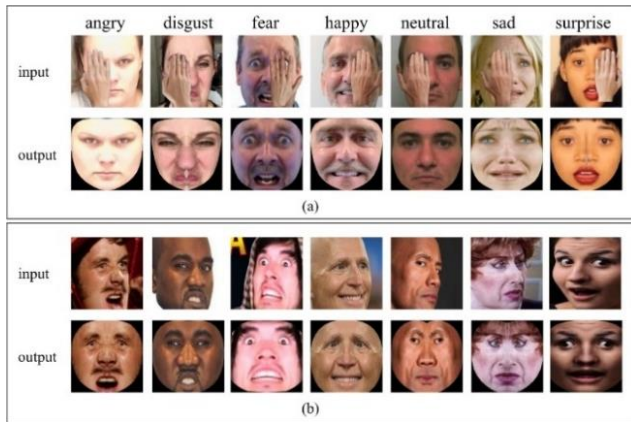


FIGURE 12. Proposed method result (a) Test result with hand-occluded face images (b) Test result with side view face images

The probabilities assigned to both the input image ("before") and output image ("after") were evaluated using the CNN model trained exclusively with frontal face data. As shown in FIGURE 13, a face labeled as "angry" in the ground truth initially exhibited a dominant model prediction of "fear" prior to pre-processing. However, after applying the proposed pre-processing method, there was a significant increase in the probability for "angry," successfully shifting the dominant emotion classification from "fear" to "angry" and thereby matching the ground truth label. The detailed probability scores for each emotion are provided in TABLE VI to substantiate these findings.

These results confirm the efficacy of the proposed approach in enhancing emotion recognition accuracy, particularly for faces affected by occlusions or non-frontal poses. By refining the input image through pre-processing, the model demonstrates an improved ability to correctly classify emotions even under challenging conditions. Moreover, this pre-processing strategy contributes to the interpretability and reliability of FER model outputs by aligning input images with the model's training distribution, reducing the likelihood of misclassification due to pose variations or occlusions. Consequently, this methodological

improvement supports more robust FER performance not only in controlled experiments but also in practical applications, where variations in facial orientation and partial occlusions are frequently encountered.

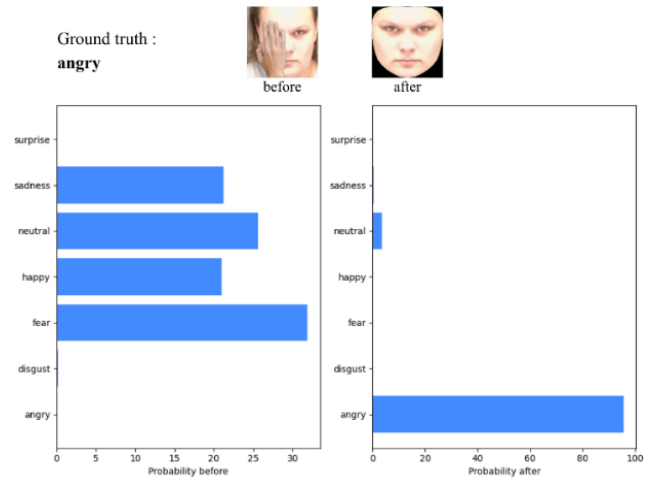


FIGURE 13. Probability of before and after proposed method implementation

FIGURE 13 illustrates the probability values for each emotion class before and after applying the proposed pre-processing method. The emotion class with the highest probability value (dominant emotion) corresponds to the output of the facial emotion recognition (FER) model, demonstrating that even a relatively small difference in probability scores can significantly impact the final classification outcome. This finding underscores the importance of precise probability adjustment in achieving reliable emotion recognition results.

The test results using the face image presented in FIGURE 13 reveal a substantial increase in the probability of the "Angry" emotion, which initially registered a value of 0.028 and subsequently rose to 95.612 after implementation of the proposed method. This dramatic shift indicates a marked improvement in emotion classification, as the dominant model prediction switched from "Fear" (Fe) to "Angry" (An), thereby aligning more closely with the ground truth label. The magnitude of the probability change (an increase of 95.584) further confirms the effectiveness of the pre-processing approach in enhancing classification accuracy for challenging cases.

To provide a more comprehensive understanding, the following section will discuss the broader impact of the proposed method on four leading FER approaches: CNN, DeepFace, FaceLib, and SVM. Performance comparisons will be conducted based on the test data depicted in FIGURE 6, with detailed quantities summarized in TABLE I, offering a robust evaluation of method generalizability and practical utility across multiple FER systems.

TABLE VI
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH CNN

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	34.00	74.00	40.00	Increased
Disgust	100	2.00	45.50	43.50	Increased
Fear	100	37.50	56.50	19.00	Increased
Happy	100	33.50	71.50	38.00	Increased
Neutral	100	2.00	61.50	59.50	Increased
Sad	100	31.00	51.50	20.50	Increased
Surprise	100	33.00	49.00	16.00	Increased
Average				33.79	Increased

TABLE VII
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH DEEPFACE

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	36.50	38.00	1.50	Increased
Disgust	100	2.00	9.00	7.00	Increased
Fear	100	26.50	26.50	0.00	No Change
Happy	100	16.00	70.00	54.00	Increased
Neutral	100	19.00	64.00	45.00	Increased
Sad	100	29.50	35.50	6.00	Increased
Surprise	100	35.50	22.00	-13.50	Decreased
Average				14.29	Increased

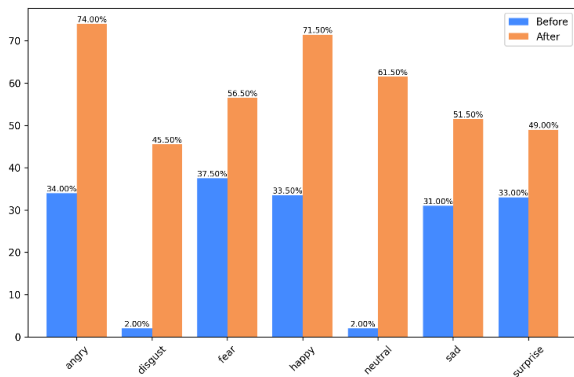


FIGURE 14. The result of implementing the proposed method on the hand-occluded face using the CNN model

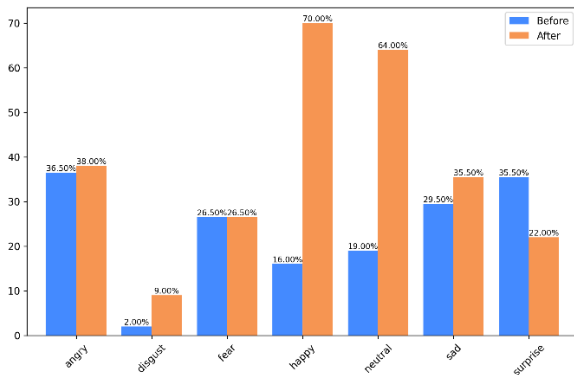


FIGURE 15. The result of implementing the proposed method on the hand-occluded face using the Deepface model

TABLE VIII
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH FACELIB

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	76.00	40.79	50.00	9.21	Increased
Disgust	88.00	3.41	3.98	0.57	Increased
Fear	99.50	48.24	65.83	17.59	Increased
Happy	99.50	45.23	77.39	32.16	Increased
Neutral	99.50	22.11	41.21	19.10	Increased
Sad	99.50	55.78	66.83	11.05	Increased
Surprise	99.50	51.26	36.18	-15.08	Decreased
Average				10.66	Increased

TABLE IX
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH SVM

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	80.50	66.00	14.50	Increased
Disgust	100	0.50	0.50	0.00	No Change
Fear	100	70.00	54.00	-16.54	Decreased
Happy	100	36.00	81.50	45.50	Increased
Neutral	100	8.00	8.50	0.50	Increased
Sad	100	12.00	45.00	33.00	Increased
Surprise	100	3.00	0.50	-2.50	Decreased
Average				6.49	Increased

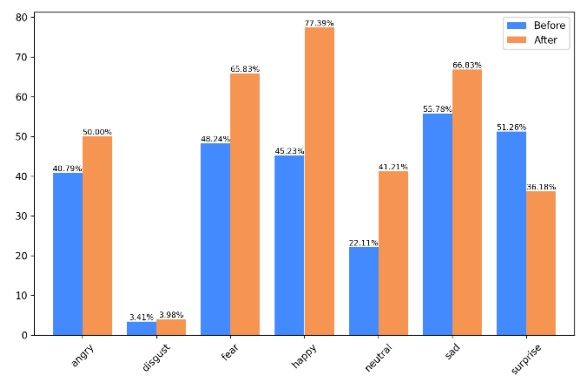


FIGURE 16. The result of implementing the proposed method on the hand-occluded face using the Facelib model

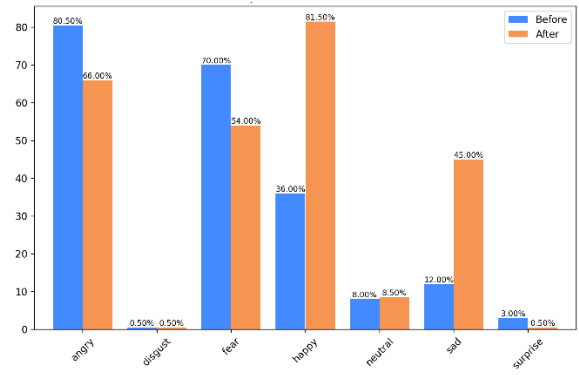


FIGURE 17. The result of implementing the proposed method on the hand-occluded face using the SVM model

TABLE X
ACCURACY RESULT OF NON-FRONTAL FACE WITH CNN

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	76.39	50.00	-26.39	Decreased
Disgust	100	2.90	14.49	11.59	Increased
Fear	100	17.35	45.92	28.57	Increased
Happy	100	18.00	32.50	14.50	Increased
Neutral	100	4.50	33.00	28.50	Increased
Sad	100	21.66	73.89	52.23	Increased
Surprise	100	27.53	34.27	6.74	Increased
Average				16.53	Increased

TABLE XI
ACCURACY RESULT OF NON-FRONTAL FACE WITH DEEPPFACE

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	34.72	36.11	1.39	Increased
Disgust	100	2.90	0.72	-2.18	Decreased
Fear	100	27.55	19.39	-8.16	Decreased
Happy	100	70.00	61.5	-8.50	Decreased
Neutral	100	42.50	56.50	14.00	Increased
Sad	100	29.50	45.22	-3.19	Decreased
Surprise	100	23.03	22.47	-0.56	Decreased
Average				-1.03	Decreased

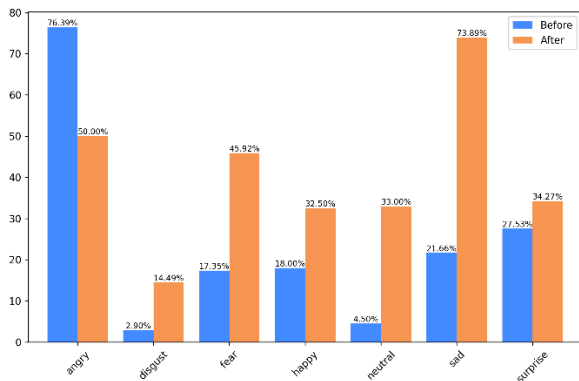


FIGURE 18. The result of implementing the proposed method on the non-frontal face using the CNN model

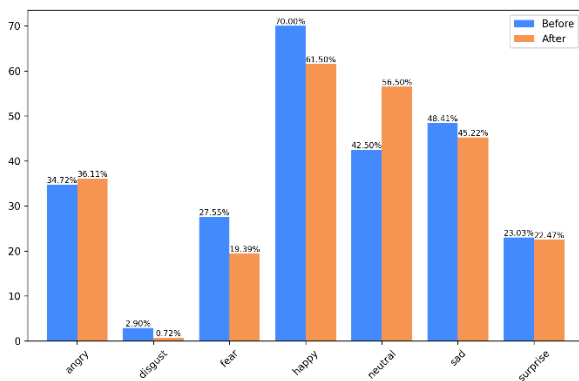


FIGURE 19. The result of implementing the proposed method on the non-frontal face using the Deepface model

TABLE XII
ACCURACY RESULT OF NON-FRONTAL FACE WITH FACELIB

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	70.83	51.11	-9.72	Decreased
Disgust	99.27	1.46	0.73	-0.73	Decreased
Fear	100	22.45	48.98	26.53	Increased
Happy	99.50	74.87	53.77	-21.10	Decreased
Neutral	99.50	46.73	31.66	-15.10	Decreased
Sad	98.08	55.19	67.63	12.44	Increased
Surprise	97.75	54.02	25.86	-28.20	Decreased
Average				-5.12	Decreased

TABLE XIII
ACCURACY RESULT OF NON-FRONTAL FACE WITH SVM

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	72.22	69.44	-2.78	Decreased
Disgust	100	0.00	0.00	0.00	No change
Fear	100	68.37	63.27	-5.10	Decreased
Happy	100	56.50	68.00	11.50	Increased
Neutral	100	14.50	5.00	-9.50	Decreased
Sad	100	14.65	38.22	23.57	Decreased
Surprise	100	3.93	1.69	-2.24	Decreased
Average				2.21	Increased

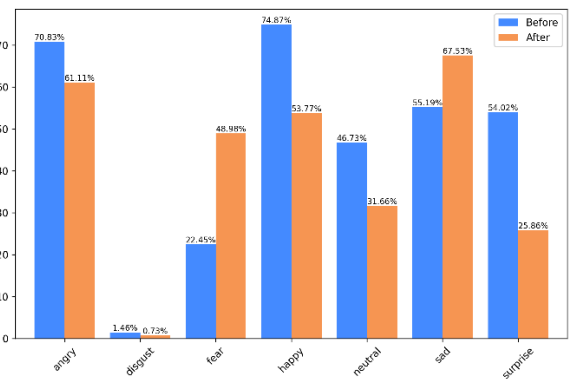


FIGURE 20. The result of implementing the proposed method on the non-frontal face using the Facelib model

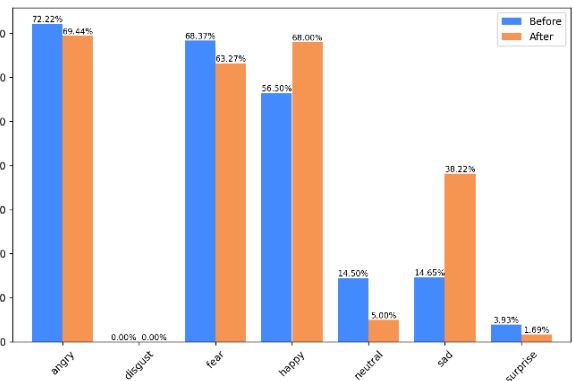


FIGURE 21. The result of implementing the proposed method on the non-frontal face using the SVM model

In addition to testing on data with vertical hand occlusion covering the face, we also evaluate the proposed method on data with horizontal hand occlusion covering the mouth area, as shown in [FIGURE 8](#). This scenario, commonly encountered in real-world situations, is tested using the carefully filtered RAF-DB dataset. Importantly, this dataset was not used in the training phase of the CNN model, which was trained solely on frontal face data. This approach ensures the model is tested on unseen data, providing an accurate assessment of its performance in real-world conditions with varying occlusions.

TABLE XIV
SELECTED RAFDB DATA DISTRIBUTION

An	Di	Fe	Ha	Ne	Sa	Su
106	62	73	200	200	200	200



FIGURE 22. The result of applying the proposed method to a face with hand occlusion in the mouth area

TABLE XV
DETECT FACE PERFORMANCE OF THE MODELS WITH HAND OCCLUSION IN THE MOUTH AREA USING SELECTED RAFDB DATA

Emotion	CNN (%)	DeepFace (%)	FaceLib (%)	SVM (%)
Angry	100	100	96.22	100
Disgust	100	100	100	100
Fear	100	100	93.15	100
Happy	100	100	99.50	100
Neutral	100	100	99.50	100
Sad	100	100	99.50	100
Surprise	100	100	99.50	100

TABLE XVII
COMPARISON RESULT

Emotion	CNN (%)		DeepFace (%)		FaceLib (%)		SVM (%)	
	A	B	A	B	A	B	A	B
Angry	-34.62	-26.39	-7.70	1.39	-14.29	-9.72	3.85	-2.78
Disgust	12.93	11.59	-3.40	-2.18	-2.04	-0.73	-0.68	0.00
Fear	17.82	28.57	-10.89	-8.16	16.83	26.53	-30.7	-5.10
Happy	29.00	14.50	-6.00	-8.50	4.53	-21.10	22.00	11.5
Neutral	28.00	28.50	17.50	14.00	-5.02	-15.07	1.00	-9.50
Sad	27.10	52.23	-15.49	-3.19	6.50	12.44	20.64	23.57
Surprise	25.34	6.74	-6.00	-0.56	-20.41	-28.16	-2.00	-2.24
Average	15.08	16.53	-4.57	-1.03	-1.99	-5.12	2.02	2.21

Positive = increase;
Negative = decrease;
A = Other method [22];
B = Our Method.

TABLE XVI
AVERAGE ACCURACY CHANGES BEFORE AND AFTER APPLYING THE METHOD ON FACES WITH HAND OCCLUSION IN THE MOUTH AREA

Emotion	CNN (%)	DeepFace (%)	FaceLib (%)	SVM (%)
Angry	-2.83	4.72	-0.98	-9.44
Disgust	-1.61	-3.23	0.00	0.00
Fear	-12.33	-5.48	14.71	0.00
Happy	-4.00	-6.50	4.02	-2.00
Neutral	32.00	-9.50	-41.70	-10.5
Sad	-21.5	-11.00	12.57	16.5
Surprise	12.00	10.5	-9.04	-0.5
Average	0.25	-2.93	-2.92	-0.85

Positive = increase;
Negative = decrease

C. COMPARISON WITH STATE-OF-THE-ART METHOD

The evaluation will also include a comparison of our method with the frontalization technique performed by K.Y. Tsai et al. (2021) [22]. They also conducted frontalization using only 2D image processing. This comparison aims to assess the relative effectiveness of our proposed method in handling facial occlusions and non-frontal faces compared to the existing frontalization approach. By benchmarking both methods, we can evaluate the strengths and weaknesses of each in terms of accuracy and robustness in real-world applications.



FIGURE 23. Comparison result, (A) input image, (B) our method, (C) other method [22]

D. ENGAGEMENT DETECTION RESULT

In this section, the results of integrating the proposed method with the FER system in online learning specifically in asynchronous learning will be presented, where student engagement is evaluated while they watch the instructional video provided by the teacher. FIGURE 24 shows a condition in which a student is watching the material presented by the instructor.

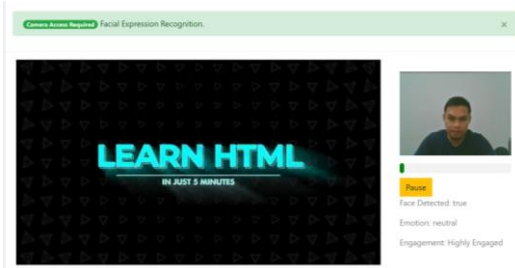


FIGURE 24. A page in asynchronous learning that contains the video material frame and the FER system frame

The system will produce an insight containing the student's engagement data, emotion percentages, and the engagement conclusion based on the dominant engagement status across the duration of the data, which corresponds to the length of the instructional video provided by the teacher. A visualization of the results can be seen in FIGURE 26.

We tested three students to evaluate their engagement levels. The first student was scripted to be predominantly bored, the second to be predominantly confused, and the third to be predominantly highly engaged (FIGURE 25).

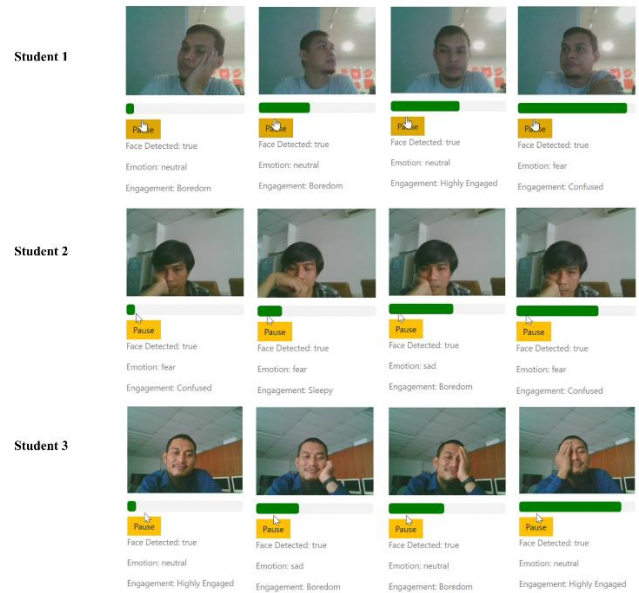


FIGURE 25. Students condition in asynchronous learning scenario

TABLE XVIII
STUDENT ENGAGEMENT COMPOSITION

Student	A	B	C	D	E
Student 1	20.00	11.11	53.33	8.89	6.67
Student 2	13.33	51.11	22.22	11.11	2.22
Student 3	51.11	2.22	35.56	6.67	4.44

A = Highly engaged; B = Confused; C = Boredom; D = Sleepy; E = Very not engaged

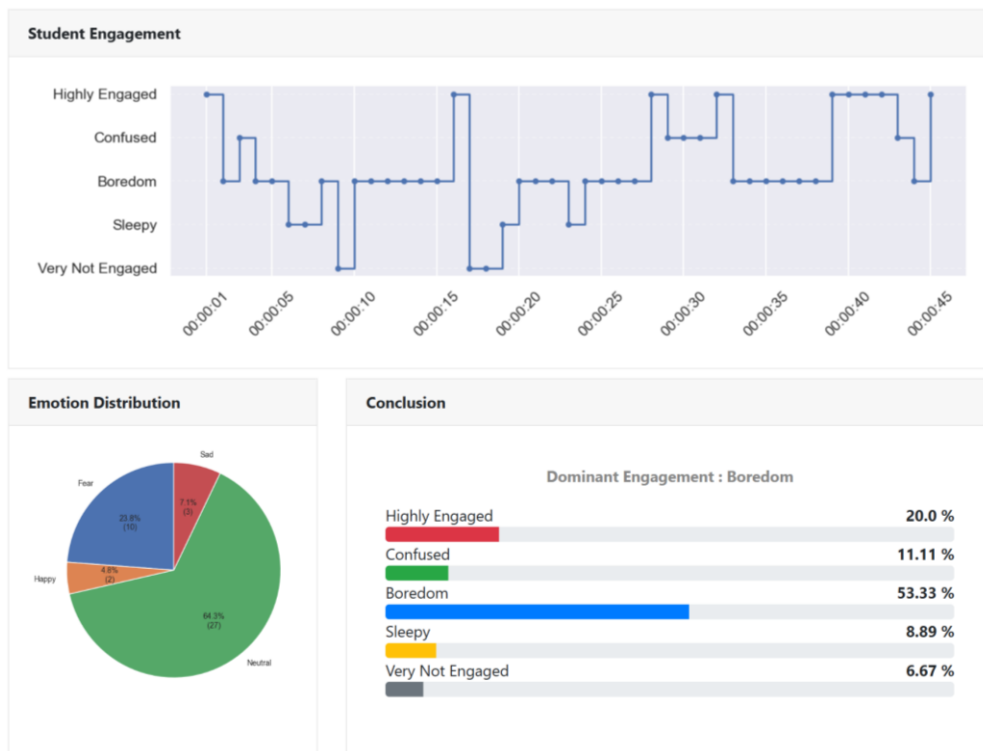


FIGURE 26. A page that contains the student's engagement insight results

VI. DISCUSSION

A. FACE DETECTOR

The first stage in a Facial Expression Recognition (FER) system is face detection. If this stage fails, the entire subsequent pipeline will stop because the system cannot extract expression features from images that are not identified as faces. According to the research conducted using four state-of-the-art methods, CNN, DeepFace, Facelib, and SVM, only the Facelib method failed in face detection, while the other three methods successfully recognized faces 100 percent of the time. Facelib uses the Single Shot Multibox Detector (SSD) algorithm for face detection.

SSD is a one-stage detector architecture that combines CNN-based feature extraction with a multi-scale feature map mechanism to detect objects of various sizes simultaneously. Its main advantage lies in its high inference speed, as both classification and bounding box regression are performed in one step. However, several studies have shown that this efficiency comes at the cost of precision, especially under conditions of occlusion or pose variation [46]. SSD performs optimally on frontal face images and those without background interference, but its performance drastically decreases when parts of the face are occluded, for example, by hands or masks, or when the face is in a non-frontal position.

The results presented in TABLE VIII and TABLE XII support this theory, where Facelib with the SSD detector achieved only 76 % performance on hand-occluded faces in the "angry" class and 88 % in the "disgust" class, while other classes, including fear, happy, neutral, sad, and surprise, showed higher results, around 99.5 %. For non-frontal face images, the detection performance for each class was 100 percent, 99.27 %, 100 %, 99.50 %, 99.50 %, 98.08 percent, and 97.75 %. The performance drop in detecting occluded faces aligns with the findings of Jang et al. (2019), who demonstrated that the Face-SSD algorithm variant can only maintain accuracy in the range of 80 to 95 percent, depending on the degree of occlusion [47].

Thus, although other studies have suggested that SSD offers superior real-time performance [48], the results in this study indicate that, without improvements in the architecture and pre-alignment stages, SSD's performance remains inadequate for FER applications in real-world environments that are characterized by occlusions and pose variations.

B. PARTIAL OCCLUSION BY HAND

The presence of hands partially covering the face is one of the most common forms of partial occlusion in real-world scenarios, such as when someone reacts spontaneously to a specific emotion or experiences boredom during online lessons, where hands are often used to support the head or face. This type of occlusion significantly reduces the performance of Facial Expression Recognition (FER) systems as it obstructs key facial areas that are crucial in determining expressions, such as the eyes, nose, and particularly the mouth. When these areas are missing from the input image, the features extracted during the process

become incomplete, making it difficult for the model to map the patterns to the correct emotion.

Experimental results clearly demonstrate this effect. On the test data before pre-processing with the proposed method (the "before" data), the average FER accuracy was only 24.71%, 23.57%, 38.12%, and 30.08% for the CNN, DeepFace, Facelib, and SVM models, respectively. These results indicate that nearly two-thirds of the classifications failed to recognize emotions correctly when the face was covered by a hand. This issue was particularly evident in deep learning-based models such as CNN and DeepFace, where despite the model's ability to learn complex features, it remained highly dependent on the completeness of the facial information.

After applying the pre-processing with the proposed method to remove the hand occlusion from the facial images, there was a significant increase in accuracy. The accuracy rose to 58.5%, 37.9%, 48.8%, and 36.6% for the four models. The average increase of over 20% demonstrates the effectiveness of the pre-processing stage in recovering important information that was previously lost. This process helps the model generate more stable feature representations, even when part of the face remains occluded by the hand.

Neutral and happy emotions were the most positively impacted by the proposed method under conditions of vertical hand occlusion. The sharp increase in accuracy for neutral emotions (up to +59.5% with CNN and +45% with DeepFace) indicates that the representation of neutral expressions greatly benefits from the proposed pre-processing in the areas obstructed by the hand. This might be due to the less prominent dynamic features of neutral expressions, such as the mouth, making it easier for the model to recognize the main structure of the face, even when partially covered. A similar sharp increase was observed for happy emotions, with DeepFace (+54%), FaceLib (+32.16%), and SVM (+45.5%). This suggests that visual indicators of happiness, such as cheek movement and contraction of the orbicularis oculi around the eyes, can still be captured by the model, even when parts of the face are covered by a hand.

On the other hand, emotions such as surprise and fear were most negatively impacted, with a notable decrease in accuracy, especially for the DeepFace and FaceLib models. This decrease indicates that the expression features for these two emotions, such as eye widening or sudden changes in the upper face, are highly susceptible to disruption due to hand occlusion on the upper portion of the face. With a hand covering parts of the eyes or forehead, the characteristic patterns of surprise and fear expressions become difficult for the model to recognize. This finding underscores the importance of the upper face in detecting highly dynamic emotions and highlights the need for developing algorithms that are more sensitive to the presence of occlusions in key expressive areas, such as the eyes and eyebrows.

From a system interpretation perspective, this improvement also indicates that a robust FER approach to occlusion requires explicit handling at the face detection or

normalization stages. In other words, the model's architecture is not the only factor determining the final performance, but also the quality and integrity of the input received. In real-world applications such as smart surveillance, driver monitoring, and emotion-based interaction, the issue of hands covering the face must be considered from the design stage of the pipeline so that the system can remain reliable, even when encountering faces that are not fully exposed.

There is a limitation in the proposed method, where the hand detection model from MediaPipe occasionally fails to detect the presence of a hand on the face, resulting in a failure in subsequent processing stages, leading to outputs that do not meet expectations (FIGURE 27). The number of hand detection failures for each emotion class was as follows: angry 9, disgust 6, fear 13, happy 17, neutral 11, and surprise 17, from a total of 200 images per emotion class. Therefore, it can be concluded that the hand detection failure rate is below 0.1%.

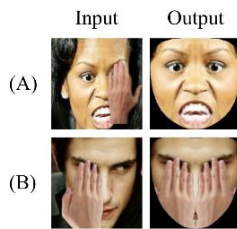


FIGURE 27. (A) Success condition (B) Fail condition

The failure of the hand detection model from MediaPipe in detecting the presence of hands on the face is generally caused by a combination of technical factors and complex visual conditions. One of the main reasons is the overlap between the hand and the face, where similar skin tones and the hand being too close to the face make it difficult for the system to distinguish the hand area from the face's contour [49]. Image quality and lighting conditions also play a crucial role, as uneven lighting, shadows on the face, or high reflections can lower the confidence score below the threshold of 0.7 set by the MediaPipe model. Failures also frequently occur when the hand size in the image is too large or too small, due to the camera distance not matching the ideal detection ratio.

C. NON-FRONTAL FACE (POSE-INVARIANT)

Another form of occlusion tested in this study is the condition of non-frontal faces or pose-invariant faces. While the pre-processing with the proposed method successfully increased the accuracy of the FER model on faces partially

occluded by hands (vertically), this pre-processing also led to a decrease in accuracy for FER models on non-frontal face conditions. The experimental results TABLE VIII – X) show that for the "before" data, the accuracy for the CNN, DeepFace, FaceLib, and SVM models were 24.05%, 35.59%, 46.51%, and 32.88%, respectively. After applying the pre-processing with the proposed method, the accuracies increased to 40.6%, 34.6%, 41.4%, and 35.1%. The positive impact was observed only in the CNN and SVM models, while the accuracy of the DeepFace and FaceLib models decreased. The increase of over 15% in the CNN model is likely due to the fact that the model was trained using only frontal face data, so the test data, which was frontal after pre-processing, was easily recognized by the model.

For DeepFace, a decrease in accuracy of -1.03% occurred, which may be due to the fact that DeepFace (FIGURE 28) already applies a frontalization pre-processing step in its architecture. As a result, when tested with the proposed method, there was a double frontalization process—first by the proposed method, then again by DeepFace. This double processing led to redundant spatial transformations, where the face, already frontalized, was processed again at the internal stage of DeepFace, causing minor deformations in the facial structure, which actually reduced classification accuracy. However, this decrease in accuracy was minimal and did not significantly affect the overall model performance. The same issue occurred with FaceLib, where FaceLib already incorporates face alignment in its pipeline, as indicated in the FaceLib Python code documentation on its official GitHub link.

The frontalization process applied in this study aims to normalize the face orientation to align with the camera's view. However, this process has the potential to cause side effects, such as texture smoothing on specific areas of the face [50]. In conditions where one side of the face is covered by a hand or there is uneven lighting distribution, this replication process can create artificial textures in those areas. On the other hand, these artificial textures may alter the authenticity of the local feature patterns around the reconstructed area, thereby affecting the image representation used by the emotion recognition system. In this context, FaceLib, which relies on spatial and textural information to extract expression features, may misinterpret these replicated patterns as characteristics of an emotion different from the actual condition. As a result, the emotion classification process may experience a degradation in accuracy due to the mismatch in texture structure resulting from the frontalization process.

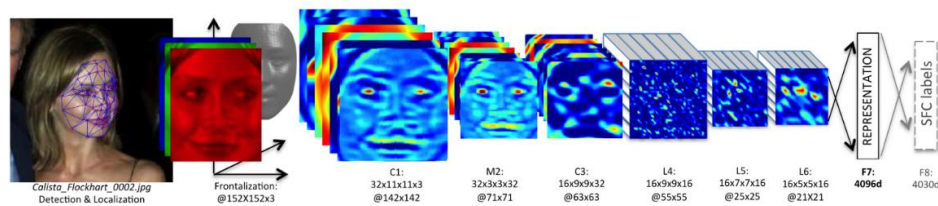


FIGURE 28. DeepFace Architecture [41]

In line with the comparison results between the proposed method and the frontalization method by K.Y. Tsai et al. (2021) [22] presented in TABLE XIV, the findings indicate that both the proposed method and the comparison method lead to similar conclusions. Specifically, there is an increase in accuracy for CNN and SVM models, while a decrease in accuracy is observed in the DeepFace and FaceLib models. As previously explained, DeepFace and FaceLib already incorporate face alignment within their pipeline, meaning that applying pre-processing before using these models can potentially lower their accuracy.

Overall, these results demonstrate that the impact of pre-processing on model performance largely depends on the design and characteristics of the architecture used. When the image correction stage aligns with the feature extraction strategy within the model, accuracy improvement can be consistently achieved. However, if the initial transformation potentially duplicates existing functions within the network, the effect can be the opposite. These findings underscore the importance of ensuring that pre-processing strategies are well-aligned with the internal design of the model to ensure optimal emotion recognition outcomes, particularly in cases of non-frontal faces.

D. PARTIAL OCCLUSION IN THE MOUTH AREA

Based on the results of the average accuracy changes before and after the application of the method on faces with hand occlusion in the mouth area, it is evident that the observed decrease in performance is relatively insignificant across most models. For instance, the CNN model only experienced a very slight increase (+0.25%), while DeepFace and FaceLib showed average decreases (-2.93% and -2.92%, respectively). SVM also exhibited a slight decline (-0.85%). It can be concluded that the proposed method did not have a significant impact on improving emotion recognition performance in the case of hand occlusion in the mouth area, and did not produce substantial differences compared to the initial condition. This indicates that the characteristic features of expressions in the mouth area, which are covered by the hand, are difficult to normalize with the approach used. As a result, the model did not gain any substantial new information for the classification process.

These results align with the phenomenon observed in facial expression recognition research when the mouth area is occluded, such as in the case of wearing medical masks. Various studies have shown that emotions such as happiness, sadness, and anger are more difficult to recognize when the mouth and nose are covered by a mask [51][52]. The decrease in accuracy can also be attributed to the loss of several important expression actions in that area, such as smiling, lip movements, or nose wrinkles, all of which contribute significantly to detecting specific emotions. In these conditions, the system tends to rely more on features from the upper face, such as the eyes and eyebrows, which are less informative for certain emotions. Moreover, this dependency often results in ambiguous emotion estimation, particularly between expressions with similar upper-face cues, such as anger and disgust. From a psychological

perspective, the mouth region provides strong and distinctive cues that humans also use to interpret emotional intent, and its absence not only challenges machine models but also human perception accuracy. Consequently, many recent studies have focused on enhancing feature extraction from the visible facial regions, integrating contextual information, or employing data augmentation techniques to compensate for the missing lower-face patterns and restore recognition performance under partial occlusion.

E. STUDENT ENGAGEMENT DURING ONLINE LEARNING

The results of this study indicate that integrating the proposed method with the FER system in an asynchronous learning scenario can provide a quantitative depiction of students' engagement patterns while watching instructional videos. The system not only identifies emotion percentages but also converts them into more applicable engagement statuses, such as highly engaged, confused, bored, sleepy, and very not engaged. Through this approach, the evaluation of student engagement becomes more measurable and objective, enabling educators to gain deeper insight into students' affective responses toward the learning material. The presence of engagement-insight visualizations further strengthens the analytical function, as educators can directly observe engagement distribution throughout the duration of the video.

The evaluation conducted on three students with different scripted behaviors demonstrates the system's capability to distinguish various engagement patterns. For example, Student 1 showed a dominance of boredom at 53.33%, Student 2 was dominated by the confused category at 51.11%, whereas Student 3 exhibited the highest level of highly engaged at 51.11%. These differences indicate that the system is able to consistently capture variations in affective expressions and behaviors, confirming the model's sensitivity to relevant emotional features. Thus, the proposed method is not only effective in detecting facial emotions but also contributes to a more comprehensive understanding of students' engagement quality in online learning.

F. FUTURE WORKS

Based on the results of the conducted research, several key areas have been identified that require further development to improve the robustness of the Facial Expression Recognition (FER) system under partial occlusion conditions. Although the proposed method provides an initial contribution in addressing hand-occluded faces, the limitations encountered, such as inconsistent effectiveness on deep learning-based models, hand detection failures in certain conditions, and minimal improvement in mouth area occlusion, indicate the need for a more adaptive and comprehensive approach. These challenges align with findings from other studies, which emphasize that facial occlusion, whether caused by masks, accessories, or hands, remains a significant barrier in computer vision-based FER systems, particularly when key facial expression features, such as the eyes, nose, and mouth, are covered.

Therefore, the development of solutions capable of addressing a wide range of occlusions, improving detection accuracy in extreme poses, and integrating multi-modal information should be a top priority for future research. These advancements are necessary to create a FER system that is truly robust in real-world applications. Several potential developments for making FER systems more resilient to various conditions are outlined below, focusing on areas such as adaptive pre-processing techniques, enhanced pose recognition models, and leveraging additional data sources like audio or physiological signals.

VII. CONCLUSION

This study shows that the proposed method yields the strongest results in cases of faces partially occluded vertically by hands, where key facial areas such as the eyes, cheeks, and eyebrows are covered. In this condition, several models, including CNN and DeepFace, exhibited clear performance improvements, with an average accuracy increase of 33.79% for CNN and 14.29% for DeepFace. Meanwhile, for faces with pose variations (non-frontal), the method also resulted in an accuracy increase, although the improvement was not significant, indicating the model's potential to adapt to different face orientations. Conversely, for faces with occlusion in the mouth area, the method did not have a meaningful impact on emotion detection accuracy, with some models, such as DeepFace (-2.93%) and FaceLib (-2.92%), actually experiencing a performance decrease. This indicates that the loss of information in the mouth area significantly affects the system's ability to interpret emotional expressions, especially for expressions like happy, sad, and fear, which generally rely heavily on lip movements.

Nevertheless, the results of this study still demonstrate that the proposed approach has the potential to enhance the robustness of expression recognition systems under certain partial occlusion conditions. Other factors, such as the failure of the MediaPipe hand detector to detect the presence of hands, also affect the overall system performance, particularly in low lighting conditions or extreme hand poses. This limitation opens the door for the development of more occlusion-adaptive models that are more sensitive to variations in hand position in the future.

It is important to note that the findings of this study are limited to the dataset used in this experiment, which has specific characteristics and image distributions. Therefore, the performance of the method may vary when tested on datasets with more diverse resolutions, more complex occlusion levels, or different visual backgrounds. Further research using multi-scenario datasets and multi-modality approaches is recommended to confirm the effectiveness of the method more broadly and expand its application to real-world scenarios.

REFERENCES

[1] R. Kimura, M. Matsunaga, E. Barroga, dan N. Hayashi, "Asynchronous e-learning with technology-enabled and enhanced training for continuing education of nurses: a scoping review," *BMC*

Med. Educ., vol. 23, no. 1, hal. 1–29, 2023, doi: 10.1186/s12909-023-04477-w.

[2] C. F. Hsu, S. P. K. Mudiyansele, R. Agustina, dan M. F. Lin, "Basic emotion detection accuracy using artificial intelligence approaches in facial emotions recognition system: A systematic review," *Appl. Soft Comput.*, vol. 172, no. June 2024, hal. 112867, 2025, doi: 10.1016/j.asoc.2025.112867.

[3] X. Zhu, Z. He, L. Zhao, Z. Dai, dan Q. Yang, "A Cascade Attention Based Facial Expression Recognition Network by Fusing Multi-Scale Spatio-Temporal Features," *Sensors*, vol. 22, no. 4, hal. 1–16, 2022, doi: 10.3390/s22041350.

[4] X. Zhang, F. Zhang, dan C. Xu, "Joint Expression Synthesis and Representation Learning for Facial Expression Recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 3, hal. 1681–1695, 2022, doi: 10.1109/TCSVT.2021.3056098.

[5] J. S. J. L. D. Freire-obregón dan M. Castrillón-santana, "Multimodal emotion recognition based on a fusion of audiovisual information with temporal dynamics," *Multimed. Tools Appl.*, 2024, doi: 10.1007/s11042-024-20227-6.

[6] D. Mamieva, A. B. Abdusalomov, A. Kutlimuratov, B. Muminov, dan T. K. Whangbo, "Multimodal Emotion Detection via Attention-Based Fusion of Extracted Facial and Speech Features," *Sensors*, vol. 23, no. 12, 2023, doi: 10.3390/s23125475.

[7] N. Ud Din, K. Javed, S. Bae, dan J. Yi, "A Novel GAN-Based Network for Unmasking of Masked Face," *IEEE Access*, vol. 8, hal. 44276–44287, 2020, doi: 10.1109/ACCESS.2020.2977386.

[8] T. Cao, C. Liu, dan J. Chen, "Nonfrontal Expression Recognition in the Wild Based on PRNet Frontalization and Muscle Feature Strengthening," *Math. Probl. Eng.*, vol. 2021, hal. 1–21, Jul 2021, doi: 10.1155/2021/6620752.

[9] A. Morar, F. Moldoveanu, A. Moldoveanu, dan V. A. Oana Balan, "GPU accelerated 2D and 3D image processing," *Proc. 2017 Fed. Conf. Comput. Sci. Inf. Syst. FedCSIS 2017*, vol. 11, hal. 653–656, 2017, doi: 10.15439/2017F265.

[10] O. S. Ekundayo dan S. Viriri, "Facial Expression Recognition: A Review of Trends and Techniques," *IEEE Access*, vol. 9, hal. 136944–136973, 2021, doi: 10.1109/ACCESS.2021.3113464.

[11] A. Sukumaran dan A. Manoharan, "Multimodal Engagement Recognition From Image Traits Using Deep Learning Techniques," *IEEE Access*, vol. 12, no. December 2023, hal. 25228–25244, 2024, doi: 10.1109/ACCESS.2024.3353053.

[12] S. N. Karimah dan S. Hasegawa, "Automatic engagement estimation in smart education/learning settings: a systematic review of engagement definitions, datasets, and methods," *Smart Learn. Environ.*, vol. 9, no. 1, 2022, doi: 10.1186/s40561-022-00212-y.

[13] S. Gupta, P. Kumar, dan R. Tekchandani, "A multimodal facial cues based engagement detection system in e-learning context using deep learning approach," *Multimed. Tools Appl.*, vol. 82, no. 18, hal. 28589–28615, 2023, doi: 10.1007/s11042-023-14392-3.

[14] N. Sun, J. Tao, J. Liu, H. Sun, dan G. Han, "3-D Facial Feature Reconstruction and Learning Network for Facial Expression Recognition in the Wild," *IEEE Trans. Cogn. Dev. Syst.*, vol. 15, no. 1, hal. 298–309, 2023, doi: 10.1109/TCDS.2022.3157772.

[15] S. Kuruvayil dan S. Palaniswamy, "Emotion recognition from facial images with simultaneous occlusion, pose and illumination variations using meta-learning," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 9, hal. 7271–7282, 2022, doi: 10.1016/j.jksuci.2021.06.012.

[16] G. Castellano, B. De Carolis, dan N. Macchiarulo, "Automatic facial emotion recognition at the COVID-19 pandemic time," *Multimed. Tools Appl.*, vol. 82, no. 9, hal. 12751–12769, 2023, doi: 10.1007/s11042-022-14050-0.

[17] S. B. Sukhavasi, S. B. Sukhavasi, K. Elleithy, A. El-Sayed, dan A. Elleithy, "Deep Neural Network Approach for Pose, Illumination, and Occlusion Invariant Driver Emotion Detection," *Int. J. Environ. Res. Public Health*, vol. 19, no. 4, 2022, doi: 10.3390/ijerph19042352.

[18] S. Bellamkonda, N. P. Gopalan, C. Mala, dan L. Settipalli, "Facial expression recognition on partially occluded faces using component based ensemble stacked CNN," *Cogn. Neurodyn.*, vol. 17, no. 4, hal. 985–1008, Agu 2023, doi: 10.1007/s11571-022-09879-y.

[19] H. Yao, X. Yang, D. Chen, Z. Wang, dan Y. Tian, "Facial Expression Recognition Based on Fine-Tuned Channel-Spatial Attention Transformer," *Sensors*, vol. 23, no. 15, hal. 6799, Jul 2023, doi:

- 10.3390/s23156799.
- [20] C. Liu, X. Liu, C. Chen, dan K. Zhou, "Deep Global Multiple-Scale and Local Patches Attention Dual-Branch Network for Pose-Invariant Facial Expression Recognition," *C. - Comput. Model. Eng. Sci.*, vol. 139, no. 1, hal. 405–440, 2023, doi: 10.32604/cmesci.2023.031040.
 - [21] Y. Tian, J. Zhu, H. Yao, dan D. Chen, "Facial Expression Recognition Based on Vision Transformer with Hybrid Local Attention," *Appl. Sci.*, vol. 14, no. 15, 2024, doi: 10.3390/app14156471.
 - [22] K. Y. Tsai, Y. W. Tsai, Y. C. Lee, J. J. Ding, dan R. Y. Chang, "Frontalization and adaptive exponential ensemble rule for deep-learning-based facial expression recognition system," *Signal Process. Image Commun.*, vol. 96, no. April, hal. 116321, 2021, doi: 10.1016/j.image.2021.116321.
 - [23] R. Khoeun, P. Chophuk, dan K. Chinnasarn, "Emotion Recognition for Partial Faces Using a Feature Vector Technique," *Sensors*, vol. 22, no. 12, 2022, doi: 10.3390/s22124633.
 - [24] K. Zheng, L. Tian, Z. Li, H. Li, dan J. Zhang, "Incorporating eyebrow and eye state information for facial expression recognition in mask-obscured scenes," *Electron. Res. Arch.*, vol. 32, no. 4, hal. 2745–2771, 2024, doi: 10.3934/ERA.2024124.
 - [25] K. Hu, G. Huang, Y. Yang, C. M. Pun, W. K. Ling, dan L. Cheng, "Rapid facial expression recognition under part occlusion based on symmetric SURF and heterogeneous soft partition network," *Multimed. Tools Appl.*, vol. 79, no. 41–42, hal. 30861–30881, 2020, doi: 10.1007/s11042-020-09566-2.
 - [26] D. Sun, W. Xie, Z. Ding, dan J. Tang, "SILP-autoencoder for face de-occlusion," *Neurocomputing*, vol. 485, hal. 47–56, 2022, doi: 10.1016/j.neucom.2022.02.035.
 - [27] M. Kolahdouzi, A. Sepas-Moghaddam, dan A. Etemad, "FaceTopoNet: Facial Expression Recognition Using Face Topology Learning," *IEEE Trans. Artif. Intell.*, vol. 4, no. 6, hal. 1526–1539, Des 2023, doi: 10.1109/TAI.2022.3207450.
 - [28] D. Poux, B. Allaert, N. Ihaddadene, I. M. Bilasco, C. Djeraba, dan M. Bennamoun, "Dynamic Facial Expression Recognition under Partial Occlusion with Optical Flow Reconstruction," *IEEE Trans. Image Process.*, vol. 31, hal. 446–457, 2022, doi: 10.1109/TIP.2021.3129120.
 - [29] Z. Zhao, Q. Liu, dan S. Wang, "Learning Deep Global Multi-Scale and Local Attention Features for Facial Expression Recognition in the Wild," *IEEE Trans. Image Process.*, vol. 30, hal. 6544–6556, 2021, doi: 10.1109/TIP.2021.3093397.
 - [30] Z. Fei, B. Zhang, W. Zhou, X. Li, Y. Zhang, dan M. Fei, "Global multi-scale extraction and local mixed multi-head attention for facial expression recognition in the wild," *Neurocomputing*, 2025, doi: 10.1016/j.neucom.2024.129323.
 - [31] Y. Huang dkk., "FERMixNet: An Occlusion Robust Facial Expression Recognition Model with Facial Mixing Augmentation and Mid-Level Representation Learning," *IEEE Trans. Affect. Comput.*, 2024, doi: 10.1109/TAFFC.2024.3454102.
 - [32] L. Xiong, J. Zhang, X. Zheng, dan Y. Wang, "Context Transformer and Adaptive Method with Visual Transformer for Robust Facial Expression Recognition," *Appl. Sci.*, vol. 14, no. 4, 2024, doi: 10.3390/app14041535.
 - [33] S. Li, W. Deng, dan J. P. Du, "Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild," *Proc. - 30th IEEE Conf. Comput. Vis. Pattern Recognition, CVPR 2017*, vol. 2017-Janua, hal. 2584–2593, 2017, doi: 10.1109/CVPR.2017.277.
 - [34] P. Lucey, J. F. Cohn, T. Kanade, J. Saragih, Z. Ambadar, dan I. Matthews, "The extended Cohn-Kanade dataset (CK+): A complete dataset for action unit and emotion-specified expression," *2010 IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. - Work. CVPRW 2010*, no. July, hal. 94–101, 2010, doi: 10.1109/CVPRW.2010.5543262.
 - [35] A. Mollahosseini, B. Hasani, dan M. H. Mahoor, "AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild," *IEEE Trans. Affect. Comput.*, vol. 10, no. 1, hal. 18–31, 2019, doi: 10.1109/TAFFC.2017.2740923.
 - [36] E. Barsoum, C. Zhang, C. C. Ferrer, dan Z. Zhang, "Training deep networks for facial expression recognition with crowd-sourced label distribution," *ICMI 2016 - Proc. 18th ACM Int. Conf. Multimodal Interact.*, hal. 279–283, 2016, doi: 10.1145/2993148.2993165.
 - [37] I. J. Goodfellow dkk., "Challenges in representation learning: A report on three machine learning contests," *Neural Networks*, vol. 64, hal. 59–63, 2015, doi: 10.1016/j.neunet.2014.09.005.
 - [38] K. Wang, X. Peng, J. Yang, D. Meng, dan Y. Qiao, "Region Attention Networks for Pose and Occlusion Robust Facial Expression Recognition," *IEEE Trans. Image Process.*, vol. 29, hal. 4057–4069, 2020, doi: 10.1109/TIP.2019.2956143.
 - [39] J. H. Cheong, E. Jolly, T. Xie, S. Byrne, M. Kenney, dan L. J. Chang, "Py-Feat: Python Facial Expression Analysis Toolbox," *Affect. Sci.*, vol. 4, no. 4, hal. 781–796, 2023, doi: 10.1007/s42761-023-00191-4.
 - [40] C. Lugaresi dkk., "MediaPipe: A Framework for Perceiving and Processing Reality," in *Third Workshop on Computer Vision for AR/VR at IEEE Computer Vision and Pattern Recognition (CVPR) 2019*, 2019. [Daring]. Tersedia pada: https://mixedreality.cs.cornell.edu/s/NewTitle_May1_MediaPipe_CVPR_CV4ARVR_Workshop_2019.pdf
 - [41] Y. Taigman, M. Yang, M. Ranzato, dan L. Wolf, "DeepFace: Closing the gap to human-level performance in face verification," *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, hal. 1701–1708, 2014, doi: 10.1109/CVPR.2014.220.
 - [42] S. Ayobi, "facelib (Version 1.2.1) [Computer software]." 2024. [Daring]. Tersedia pada: <https://pypi.org/project/facelib/>
 - [43] P. Szczakowska, A. Wosiak, dan K. Zykwinska, "Improving Automatic Recognition of Emotional States Using EEG Data Augmentation Techniques," *Procedia Comput. Sci.*, vol. 225, hal. 4225–4234, 2023, doi: 10.1016/j.procs.2023.10.419.
 - [44] C. Shorten dan T. M. Khoshgoftaar, "A survey on Image Data Augmentation for Deep Learning," *J. Big Data*, vol. 6, no. 1, 2019, doi: 10.1186/s40537-019-0197-0.
 - [45] D. A. Pitaloka, A. Wulandari, T. Basaruddin, dan D. Y. Liliana, "Enhancing CNN with Preprocessing Stage in Automatic Emotion Recognition," *Procedia Comput. Sci.*, vol. 116, hal. 523–529, 2017, doi: 10.1016/j.procs.2017.10.038.
 - [46] Z. Qu, X. Shang, S. F. Xia, T. M. Yi, dan D. Y. Zhou, "A method of single-shot target detection with multi-scale feature fusion and feature enhancement," *IET Image Process.*, vol. 16, no. 6, hal. 1752–1763, Mei 2022, doi: 10.1049/ipr2.12445.
 - [47] Y. Jang, H. Gunes, dan I. Patras, "Registration-free Face-SSD: Single shot analysis of smiles, facial attributes, and affect in the wild," *Comput. Vis. Image Underst.*, vol. 182, no. July 2018, hal. 17–29, 2019, doi: 10.1016/j.cviu.2019.01.006.
 - [48] S. Tariyal, R. Chauhan, Y. Bijalwan, R. Rawat, dan R. Gupta, "A comparative study of MTCNN, Viola-Jones, SSD and YOLO face detection algorithms," in *2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE)*, 2024, hal. 1–7. doi: 10.1109/IITCEE59897.2024.10467445.
 - [49] M. Pu, C. Y. Y. Chong, dan M. K. K. Lim, "Robustness Evaluation in Hand Pose Estimation Models using Metamorphic Testing," in *2023 IEEE/ACM 8th International Workshop on Metamorphic Testing (MET)*, IEEE, Mei 2023, hal. 31–38. doi: 10.1109/MET59151.2023.00012.
 - [50] T. Hassner, S. Harel, E. Paz, dan R. Enbar, "Effective face frontalization in unconstrained images," *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 07-12-June, hal. 4295–4304, 2015, doi: 10.1109/CVPR.2015.7299058.
 - [51] M. Grahlow, C. I. Rupp, dan B. Derntl, "The impact of face masks on emotion recognition performance and perception of threat," *PLoS One*, vol. 17, no. 2, hal. e0262840, Feb 2022, doi: 10.1371/journal.pone.0262840.
 - [52] L. Ruan, Y. Han, J. Sun, Q. Chen, dan J. Li, "Facial expression recognition in facial occlusion scenarios: A path selection multi-network," *Displays*, vol. 74, hal. 102245, 2022, doi: 10.1016/j.displa.2022.102245.



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